Binary Scaling and the Accelerated Collatz Map

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Abstract

This note studies the elementary inequality $2^n > 3^{n/2}$ alongside the accelerated Collatz map. Every odd starting value greater than 1 and congruent to 1 modulo 8 has an explicit finite decreasing segment whose length is determined solely by its lowest positive binary exponent; the family $2^x + 1$ is a special case. Beyond these forced segments, the same type of reasoning becomes heuristic for general sparse or dense binary expansions because the valuation of $3n + 1$ is global and changes from step to step. A random valuation model predicts negative logarithmic drift, but neither that model nor the inequality provides uniform descent for every orbit. Motivated by Stérin and Woods' binary-ternary automaton, an exact joint transition law for $v_2(3n + 1)$ and $v_3(n + 1)$ identifies one genuine escape mechanism and also shows why these two valuations do not form a closed global state. It yields arbitrarily long exact (2, 0) excursions and an exact descent criterion for an expanding run followed by a valuation-2 recovery block. The note separates the exact calculation from these heuristic extensions and identifies the point at which a proof of the Collatz conjecture would require new information. An exact dictionary with a cited inverse-history construction yields a positive-density logarithmic-time corollary while leaving the pointwise exceptional set uncontrolled. Combining Tao's almost-boundedness theorem with an elementary return calculation shows that, for every prescribed divergent bound F, logarithmically almost every odd orbit contains a value $q < F(N)$ with q congruent to 1 modulo 4 and hence an immediate local descent certificate. A bounded-horizon theorem shows that any positive surplus forces descent before a fixed fraction of the logarithmic size horizon, once the start is sufficiently large. This reduces a hypothetical non-descending prefix to a critical-line residue problem and isolates a concrete growth condition on its least positive representatives. A common-modulus Fourier analysis gives exact recursions for the union of all noncrossing cylinders. Its finite computations show low-frequency dispersion rather than the one-sided repulsion required by the representative-growth criterion; ordinary equidistribution would predict exponentially many survivors below the critical $(3/2)^r$ scale, not their absence. An adelic generating-function argument excludes every permanently subcritical valuation word whose weighted generating function is rational, including every eventually periodic subcritical word. A complementary band-growth estimate excludes all infinite valuation words with cumulative excess $h_r = K_r - r$ of order $o(\log r)$, while leaving a genuinely aperiodic corridor open. A d-kernel analysis shows that Collatz admissibility does not supply the finite-dimensional regularity needed for an automatic or Mahler-function argument: even the fixed orbit at 1 has an infinite-dimensional d-kernel for every base d. Explicit aperiodic 2-adic models exhibit both regular and nonregular behavior, confining any regular counterexample model to a narrow logarithmic excess band without proving that such a model comes from a positive integer. Exact counts and arbitrary-precision experiments from 100 to 1000 bits explain why ordinary random starts are exponentially repelled from the noncrossing corridor while positive truncations of negative 2-adic periodic points can shadow it for thousands of steps. They reduce the remaining pointwise issue to

divergence of the least positive representative L(r), not merely decay of the survivor density. A companion Lean file machine-checks a selected arithmetic core without asserting any global convergence theorem.

Keywords Collatz conjecture, accelerated Collatz map, binary expansion, 2-adic valuation, 3-adic valuation, base conversion, d-regular sequences, heuristic drift

1 Introduction

For a positive odd integer n, the accelerated Collatz map applies the odd Collatz operation and then removes every factor of 2. Writing $v_2(m)$ for the exponent of 2 in a positive integer m, define

$$T(n) = (3n + 1) / 2^{(v_2(3n + 1))}.$$

The Collatz conjecture is equivalent to the assertion that repeated application of the ordinary map sends every positive integer to 1. On odd starting values, the accelerated map records the same trajectory after suppressing the intervening even terms.

Binary notation suggests comparing powers of 2 generated by the size of an integer with powers of 3 generated by repeated odd steps. That comparison is exact during a forced initial segment for every odd integer congruent to 1 modulo 8. For a general later iterate, however, the successive valuations $v_2(3n + 1)$ determine whether the orbit grows or contracts. The distinction between these two settings is the main subject of this note.

2 An elementary exponential inequality

The numerical comparison used in the binary scaling picture is the following.

Proposition 1 For every positive integer n,

$$2^n > 3^{(n/2)}, \quad \text{equivalently} \quad 2^{(2n)} > 3^n.$$

Proof Since $4 > 3$, raising both sides to the positive integer power n gives $4^n > 3^n$. Because $4^n = 2^{2n}$, this is the integer-exponent form of the claim. Taking positive square roots gives $2^n > 3^{n/2}$. □

The inequality is correct for every positive integer. Its relevance to a Collatz orbit depends on whether the orbit has first been reduced to an expression to which the comparison legitimately applies. The inequality does not permit the independent evolution of the terms in an arbitrary binary expansion.

3 Exact calculation for the family 2^x plus 1

The family $n = 2^x + 1$ has a controlled initial orbit because the valuation is exactly 2 while the remaining exponent is at least 3. This produces a finite recurrence with no probabilistic assumption.

Proposition 2 Let $x \geq 3$ and set $J = \lfloor (x - 1)/2 \rfloor$. Then

$$T^j(2^x + 1) = 3^j 2^{(x - 2j)} + 1, \quad 0 \leq j \leq J.$$

Every transition described by this recurrence uses exactly two divisions by 2 while the current exponent $x - 2j$ is at least 3. In particular, the starting value reaches a smaller positive integer after one accelerated step.

Proof For the first step,

$$3(2^x + 1) + 1 = 3 \cdot 2^x + 4 = 4(3 \cdot 2^{(x-2)} + 1).$$

When $x \geq 3$, the factor in parentheses is odd, so the valuation is exactly 2 and $T(2^x + 1) = 3 \cdot 2^{x-2} + 1$. Suppose the displayed recurrence holds at index j and $x - 2j \geq 3$. Applying the same calculation replaces $3^j 2^{x-2j}$ by $3^{j+1} 2^{x-2j-2}$ and preserves the final +1. This proves the recurrence through $j = J$.

$$3^j 2^{(x-2j)} = 2^x (3/4)^j < 2^x \quad \text{for } 1 \leq j \leq J.$$

The quantity on the left is an integer, so adding 1 gives $T^j(2^x + 1) \leq 2^x < 2^x + 1$. For $j = 1$ this already proves strict descent from the starting value. □

The recurrence stops when the remaining exponent reaches 1 or 2 because the next valuation need not equal 2. Nothing in Proposition 2 supplies a closed formula for the later orbit. The separate case $x = 2$ gives $T(5) = 1$.

4 One step behavior by residue class

The residue of an odd integer modulo 4 determines whether one accelerated step contracts or expands. This observation is elementary and does not require Proposition 1.

Integers congruent to 1 modulo 4

If $n \equiv 1 \bmod 4$, then $3n + 1$ is divisible by 4, so $v_2(3n + 1) \geq 2$. For $n > 1$,

$$T(n) \leq (3n + 1)/4 < n.$$

Thus this residue class decreases in one accelerated step.

Integers congruent to 3 modulo 4

If $n \equiv 3 \bmod 4$, then $3n + 1 \equiv 2 \bmod 4$. Hence $v_2(3n + 1) = 1$ and

$$T(n) = (3n + 1)/2.$$

This value exceeds n for every $n > 1$ and is approximately $3n/2$. Its leading binary exponent can increase by one. Later valuations determine whether this growth is eventually reversed, and Proposition 1 does not control their sequence.

Refinement modulo 8

Modulo 8 separates the contracting class $n \equiv 1 \bmod 4$ into two more precise cases. If $n = 8q + 1$, then

$$3n + 1 = 4(6q + 1).$$

The cofactor $6q + 1$ is odd, so $v_2(3n + 1) = 2$ exactly. For $q > 0$, the accelerated value $6q + 1$ is smaller than $8q + 1$. If $n = 8q + 5$, then

$$3n + 1 = 8(3q + 2).$$

Hence $v_2(3n + 1) \geq 3$. Division by the displayed factor 8 already gives $3q + 2 < 8q + 5$; any additional divisions make the accelerated value smaller still. The parity of $3q + 2$ decides whether the valuation is exactly 3 or larger, illustrating why a fixed modulus may provide either an exact value or only a lower bound.

Refinement modulo 16

Modulo 16 gives the following complete refinement for odd residues. For n = 16q + 1 and n = 16q + 9,

$$3(16q + 1) + 1 = 4(12q + 1), \quad 3(16q + 9) + 1 = 4(12q + 7).$$

Both cofactors are odd, so the valuation is exactly 2. For n = 16q + 13,

$$3(16q + 13) + 1 = 8(6q + 5),$$

and the odd cofactor shows that the valuation is exactly 3. For n = 16q + 5,

$$3(16q + 5) + 1 = 16(3q + 1),$$

so the valuation is at least 4; its exact value requires finer residue information. Each of these four classes decreases in one accelerated step. The remaining odd classes 3, 7, 11, and 15 modulo 16 are all congruent to 3 modulo 4, so their valuation is exactly 1 and they increase. For n ≡ 9 mod 16, the numerator is 12 mod 16 rather than 4 mod 16, but the conclusion $v_2(3n + 1) = 2$ is unchanged.

5 Sparse binary expansions

Write an odd integer as $n = 2^a + M$, where $0 < M < 2^a$ and M is odd. Under one accelerated step,

$$T(n) = (3 \cdot 2\text{^}a + 3M + 1) / 2\text{^}k, \quad k = v_2(3n + 1).$$

If k ≥ 2, the whole integer decreases. If k = 1, the scale of the leading term is multiplied by approximately 3/2 and the leading exponent may increase. When the binary support is sparse and widely separated, it is natural to track the leading contribution as though distant lower terms had limited influence. That interpretation can be useful for constructing examples and statistical models, but it is not a component-wise form of the Collatz map.

A sparse starting expansion need not remain sparse, and sparsity does not prescribe the later valuations. A net decrease may be expected only if the evolving expansion remains suitably controlled and the accumulated powers of 2 outweigh the accumulated powers of 3. Neither condition is known to hold uniformly for all sparse starting values.

In particular, separation between the initial 1-bits is not by itself an invariant or a theorem about later steps. Multiplication by 3 and the additive term can alter several adjacent bits, and the subsequent division is governed by the low-order residue of the entire integer. Accordingly, no component-wise rule or uniform controlled-segment length follows merely from a lower bound on the initial gaps.

Two further sparse families

The family $n = 2^x + 3$ with x ≥ 4 demonstrates that two nonzero high-level binary blocks do not guarantee immediate descent. Its numerator factors as

$$3(2\text{^}x + 3) + 1 = 2(3 \cdot 2\text{^}(x - 1) + 5).$$

The cofactor is odd, so the valuation is exactly 1, and $3 \cdot 2^{x-1} + 5 > 2^x + 3$. The first accelerated step therefore increases every member of this family.

A contrasting family is $n = 2^x + 2^y + 1$ with x > y ≥ 3. Here

$$3n + 1 = 4(3 \cdot 2\text{^}(x - 2) + 3 \cdot 2\text{^}(y - 2) + 1).$$

The factor in parentheses is odd, so the valuation is exactly 2 and

$$T(n) = 3 \cdot 2^{(x-2)} + 3 \cdot 2^{(y-2)} + 1 < 2^x + 2^y + 1.$$

The restriction $y \geq 3$ is essential for this formula. When $y = 2$, the valuation is at least 4 rather than 2. No lower bound on the gap $x - y$ is needed. Further controlled steps are possible while the smallest remaining exponent is at least 3.

Gap independent controlled segment

The preceding calculation extends to every odd integer greater than 1 whose lowest positive binary exponent is at least 3. Write

$$n_0 = 2^y m + 1, \quad y \geq 3, \quad m \geq 1.$$

If y is the lowest positive exponent then m is odd, although oddness of m is not needed for the algebra below. Set $J = \lfloor (y-1)/2 \rfloor$. For $0 \leq j \leq J$,

$$T^j(n_0) = 3^j 2^{(y-2j)} m + 1.$$

Thus the odd factor from the higher binary part does not need a step-dependent notation: the same odd integer m works for every j, while the coefficient 3^j accounts for its densification in ordinary binary notation.

Indeed, whenever $y - 2j \geq 3$,

$$3(3^j 2^{(y-2j)} m + 1) + 1 = 4(3^{(j+1)} 2^{(y-2j-2)} m + 1),$$

and the cofactor is odd. Thus the valuation is exactly 2 and the recurrence continues. Moreover, for every $1 \leq j \leq J$,

$$T^j(n_0) = (3/4)^j (n_0 - 1) + 1 < n_0.$$

This theorem depends only on the lowest positive exponent, not on the gaps between higher 1-bits and not on the Hamming weight. Equivalently, it applies to every odd $n_0 > 1$ with $n_0 \equiv 1 \bmod 8$. Every two-block, three-block, four-block, or arbitrary finite-support family is therefore covered whenever its lowest positive exponent is at least 3. Multiplication by 3 may increase the ordinary binary Hamming weight, but that binary densification does not break the displayed algebraic recurrence. The recurrence ends only when its remaining power of 2 is too small to force valuation 2.

Formal binary support and the controlled segment

Let $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. For every positive integer O define its binary support by

$$E_o = \{e \in \mathbb{N}_0 \mid \lfloor O/2^e \rfloor \equiv 1 \bmod 2\}.$$

This is the unique finite nonempty set satisfying the binary reconstruction formula

$$O = \sum 2^e, \quad e \in E_o.$$

In particular, O is odd exactly when $0 \in E_o$, and its Hamming weight is $|E_o|$. For an odd O, order the support as $E_o = \{0 = e_0 < e_1 < \cdots < e_{s-1}\}$; its successive gaps are $g_i = e_i - e_{i-1}$.

Suppose $e_1 \geq 3$. Direct expansion, with no hypothesis on the higher gaps, gives

$$3O + 1 = 4(1 + \sum 3 \cdot 2^{(e_i - 2)}), \quad 1 \leq i < s.$$

The cofactor is odd, so $v_2(3O + 1) = 2$ and

$$T(O) = 1 + \sum 3 \cdot 2^{(e_i - 2)}, \quad 1 \leq i < s.$$

Because $3\cdot 2^a = 2^a + 2^{a+1}$, each positive input bit normally becomes a two-bit block. If all successive positive gaps are at least 2, these blocks are disjoint and the first-step support is

$$E(T(O)) = \{0\} \cup \bigcup \{e_i - 2, e_i - 1\}, \quad 1 \le i < s.$$

Thus the proposed rule $E(T(O)) = \{0\} \cup \{e_i - 2\}$ is not correct, even under gaps of at least 3. For example, $E_9 = \{0, 3\}$, but $T(9) = 7$ and $E_7 = \{0, 1, 2\}$. The Hamming weight therefore need not remain fixed. A gap bound of 3 is sufficient for the displayed disjoint-block formula but is not sharp; a bound of 2 already suffices for one step.

The exact finite recurrence is nevertheless simpler and stronger than a support-shift rule. Let $J = \lfloor (e_1 - 1)/2 \rfloor$. Then, for $0 \le j \le J$,

$$T^j(O) = 1 + 3^j \sum 2^{(e_i - 2j)} = 1 + (3/4)^j(O - 1), \quad 1 \le i < s.$$

For every $1 \le j \le J$ this is strictly smaller than O. The formula depends only on the lowest positive exponent e_1; neither the minimal gap nor the Hamming weight affects its validity. At the level of ordinary binary support, let $E(3^j)$ denote the support of 3^j. When the translated sets $E(3^j) + (e_i - 2j)$ are pairwise disjoint, $E(T^j(O))$ is their union together with $\{0\}$. If they overlap, binary carries merge the translated blocks, but the algebraic recurrence remains exact.

The valuation bookkeeping inside this segment is also exact. With $K_r = k_0 + \cdots + k_{r-1}$, $M_r = 2^{(K_r)}/3^r$, and

$$Q(r) = \sum 3^{(r - j)} / 2^{(K(r) - K(j))}, \quad 0 \le j < r,$$

one has, for $1 \le r \le J$,

$$K(r) = 2r, \quad M(r) = (4/3)^r, \quad Q(r) = 3(1 - (3/4)^r) < 3.$$

Accordingly the exact descent criterion $M(r)(1 - Q(r)/(3O)) > 1$ holds throughout the nonempty controlled segment. At its endpoint $e_1 - 2J$ is 1 or 2. The following valuation need not be 2, the support may change in a more complicated way, and the forced recurrence stops.

The same notation describes trailing runs cleanly: O ends in exactly $\ell \ge 2$ binary ones precisely when $\{0, 1, \ldots, \ell - 1\} \subseteq E_o$ and $\ell \notin E_o$. Section 6 shows that this condition forces $v_2(3O + 1) = 1$ and a first-step increase. The support-set formulation therefore unifies the local statements, but it does not control the later sequence of supports after the forced segment ends.

6 Trailing runs of binary ones

Long trailing runs illustrate why an informal carry argument can be misleading. Suppose an odd integer has exactly $\ell \ge 2$ trailing binary ones. Then it has the unique representation

$$n = 2^{\ell} q + (2^{\ell} - 1).$$

Here q is even and nonnegative; if q were odd, the trailing run would contain more than ℓ ones. A direct calculation gives

$$3n + 1 = 2(3 \cdot 2^{(\ell - 1)}(q + 1) - 1).$$

The expression in parentheses is odd, so $v_2(3n + 1) = 1$ and

$$T(n) = 3 \cdot 2^{(\ell - 1)}(q + 1) - 1.$$

For $n > 1$ this value is larger than n, as also follows from $n \equiv 3 \bmod 4$ and Section 4. Thus the trailing pattern gives a complete description of the first accelerated step: it forces exactly one division by 2 and

a strict increase. It neither creates a large immediate valuation nor determines any later valuation. In fact, proving convergence for every integer in this class would prove the full Collatz conjecture: all odd integers congruent to 3 modulo 4 lie in the class, while every odd integer congruent to 1 modulo 4 immediately descends and can be handled by strong induction.

7 Random valuation model

A standard heuristic treats the low binary digits of a large odd integer as independent fair bits. In that model the valuation k has the geometric distribution

$$\Pr(k = j) = 2^{(-j)}, \quad j \geq 1,$$

and therefore E[k] = 2. Ignoring the additive term at large scales, one accelerated step has multiplier $3/2^k$. The expected logarithmic change is

$$E[\log_2(3/2^k)] = \log_2 3 - 2 = \log_2(3/4) < 0.$$

The geometric mean multiplier is consequently 3/4, which predicts negative logarithmic drift. The arithmetic mean tells a different story:

$$E[3/2^k] = 3\,\Sigma_{(j \geq 1)}\, 4^{(-j)} = 1.$$

This difference reflects the influence of uncommon larger excursions on an arithmetic average. More importantly, the model does not prove that valuations along a fixed orbit behave independently. Negative average logarithmic drift in a random model cannot be promoted to a statement about every deterministic trajectory without an additional uniform argument.

8 Exact finite iterate formula

The accumulated additive terms can be written explicitly for any finite accelerated trajectory. Let $n_0, n_1, \ldots, n_r$ be odd integers satisfying

$$2^{(k_i)}\, n_{(i+1)} = 3n_i + 1, \quad 0 \leq i < r,$$

where $k_i = v_2(3n_i + 1)$. Define $K_0 = 0$ and $K_j = k_0 + \cdots + k_{j-1}$ for $j \geq 1$. Repeated substitution gives the exact identity

$$2^{(K_r)}\, n_r = 3^r n_0 + \Sigma_{(j=0)}^{(r-1)}\, 3^{(r-1-j)}\, 2^{(K_j)}.$$

The identity follows by induction on r. The step from r to r + 1 multiplies the previous equation by 3 and adds 2^{K_r} before increasing the total denominator exponent from K_r to K_{r+1}. For two steps it becomes

$$2^{(k_0 + k_1)}\, n_2 = 9n_0 + 3 + 2^{(k_0)}.$$

This formula isolates the exact obstruction to a purely multiplicative argument. Let

$$A_r = \Sigma_{(j=0)}^{(r-1)}\, 3^{(r-1-j)}\, 2^{(K_j)}.$$

Then $n_r < n_0$ holds exactly when

$$A_r < (2^{(K_r)} - 3^r)n_0.$$

In particular, the inequality $2^{K_r} > 3^r$ is necessary for this comparison, but it is not sufficient: the positive correction A_r must also fit inside the multiplicative margin. The valuation sequence controls both K_r and every exponent appearing in A_r. The general Collatz problem is therefore not reduced to the sign of a single product.

Exact descent certificates

Normalize the multiplicative surplus and the accumulated additive burden by setting

$$M_r = 2\^(K_r)/3\^r, \quad Q_r = 3A_r/2\^(K_r).$$

$$Q_r = \Sigma_(j = 0)\^(r - 1)\ 3\^(r - j)/2\^(K_r - K_j).$$

The finite-iterate identity rearranges to

$$n_r < n_0 \quad \Leftrightarrow \quad M_r(1 - Q_r/(3n_0)) > 1.$$

Because M_r is positive, this is equivalent to the pair of inequalities

$$Q_r < 3n_0 \quad \text{and} \quad M_r > 1/(1 - Q_r/(3n_0)).$$

Thus the search for a later value below n_0 is exactly the search for an index r satisfying this certificate. Several local cases supply such an index immediately.

Case A Values congruent to 1 modulo 4. For $n_0 > 1$ in this residue class, $k_0 \geq 2$ and $T(n_0) < n_0$. Hence r = 1 satisfies the certificate.

Case B The controlled segment. If the lowest positive binary exponent e_1 is at least 3, then $k_i = 2$ throughout the controlled segment. For $1 \leq r \leq J$,

$$M_r = (4/3)\^r, \quad Q_r = 3(1 - (3/4)\^r) < 3.$$

The certificate becomes

$$(4/3)\^r(1 - (1 - (3/4)\^r)/n_0) > 1,$$

which holds for every $r \geq 1$ when $n_0 > 1$. Every index inside the controlled segment therefore works, including r = 1. The segment is nonempty exactly for odd $n_0 > 1$ with $n_0 \equiv 1 \bmod 8$.

Case C An exact trailing run of at least two ones. The first valuation is $k_0 = 1$, so the first accelerated step increases the integer and r = 1 does not work. The image may enter Case A or Case B, but the length of the trailing run alone supplies no uniform bound on when this happens.

Orbit search as a semialgorithm

The normalizations can be updated without recomputing the finite sums. If the next valuation is k_r, then

$$M(r + 1) = M(r) \cdot 2\^(k_r)/3, \quad Q(r + 1) = (3/2\^(k_r))(Q(r) + 1).$$

This gives the following exact orbit search.

```
r <- 0;  n <- n0;  M <- 1;  Q <- 0
repeat
    k <- v2(3n + 1)
    n <- (3n + 1) / 2^k
    M <- M * (2^k / 3)
    Q <- (3 / 2^k) * (Q + 1)
    r <- r + 1
    if Q < 3*n0 and M > 1/(1 - Q/(3*n0)) then return r
```

For any starting value whose orbit is known to fall below its start, the loop terminates at the first such iterate. It is therefore a semidecision procedure for eventual descent. Proving that it terminates for every odd $n_0 > 1$ is not a separate simplification of the Collatz problem: it is precisely universal eventual descent, and hence is equivalent to the conjecture.

Surplus records and universal termination

The descent certificate has a useful weighted-record form. Put

$$\alpha = \log_2 3, \quad S_r = K_r - r\alpha,$$

$$P_r = \Sigma_{j=0}^{r-1} 2^{S_j}.$$

Then $M_r = 2^{sr}$, and the definition of Q_r gives

$$Q_r = \Sigma_{j=0}^{r-1} 2^{S_j - S_r} = P_r / 2^{S_r}.$$

Substitution into the finite-iterate identity yields the exact ratio

$$n_r / n_0 = 2^{-S_r}\left(1 + P_r/(3n_0)\right).$$

Consequently, descent occurs exactly when

$$n_r < n_0 \quad \Leftrightarrow \quad S_r > \log_2\left(1 + P_r/(3n_0)\right).$$

Thus S_r must be a sufficiently strong record relative to the exponentially weighted history P_r. A large total surplus is not enough if much of it was accumulated early and a later expanding stretch has made the historical burden large.

Necessary prefix bounds

If the semialgorithm never halts, then for every r

$$2^{S_r} \le 1 + P_r/(3n_0).$$

Because $P_{r+1} = P_r + 2^{sr}$, this implies

$$P_{r+1} \le \left(1 + 1/(3n_0)\right)P_r + 1.$$

Induction on this scalar recurrence gives

$$P_r \le 3n_0\left[\left(1 + 1/(3n_0)\right)^r - 1\right],$$

$$S_r \le r \log_2\left(1 + 1/(3n_0)\right),$$

$$K_r \le r \log_2\left(3 + 1/n_0\right).$$

The first bound is optimal for the scalar recurrence inequality; it is not asserted to be attained by a genuine Collatz orbit with $n_0 > 1$. A counterexample would have to respect the valuation bound at every prefix, not merely in an asymptotic average.

Cycle or divergence

Suppose an orbit never falls below n_0. If a value repeats, determinism produces a nontrivial cycle. If no value repeats, every finite set is visited only finitely often, so n_r tends to infinity. The exact one-step ratios give

$$\log_2(n_r/n_0) = r\alpha - K_r + \Sigma_{i=0}^{r-1} \log_2\left(1 + 1/(3n_i)\right).$$

In the divergent case the summands in the correction tend to zero, so their Cesàro average tends to zero. Any divergent counterexample must therefore satisfy

$$\limsup K_r/r \le \log_2 3 \quad \text{as } r \to \infty.$$

A sufficient global theorem would exclude nontrivial cycles and prove lim sup $K_r/r > \log_2 3$ for every nonperiodic positive orbit. The conjunction implies universal descent, and the Collatz conjecture

implies both statements, so the conjunction is equivalent to the conjecture. Neither statement alone is sufficient: cycle exclusion leaves open divergence, while a theorem restricted to nonperiodic orbits says nothing about periodic ones.

Finite modular patterns

Assume the current value undergoes exactly t consecutive valuations equal to one. Then

$$n + 1 = 2^{(t+1)}u \quad \text{with } u \text{ odd},$$

$$n_j = 3^j(n+1)/2^j - 1 \quad (0 \le j \le t).$$

At the end of the run,

$$n_t = 2 \cdot 3^t u - 1,$$

$$k_t = 1 + v_2(3^{(t+1)}u - 1).$$

For every t there are odd choices of u for which $k_t = 2$. An arbitrarily long expanding run therefore need not be followed by a large compensating valuation, and no recovery bound can depend only on the preceding run length.

More generally, every finite vector of positive valuations $(k_0, \ldots, k_{r-1})$ is realized by one odd residue class modulo 2^{K_r+1}, hence by infinitely many positive starting values. The nested classes attached to an infinite valuation word determine a unique 2-adic odd integer. The constant word (1, 1, ...) gives the negative fixed point −1, while the periodic word (1, 2, 1, 2, ...) gives the negative cycle −5, −7. Finite modular arithmetic therefore cannot forbid all low-valuation paths; a global proof must use the Archimedean fact that the start is an ordinary positive integer.

Binary and ternary structure

Stérin and Woods construct a two-dimensional quasi-cellular automaton in which horizontal binary data and vertical ternary data describe the same Collatz computation. Its two finite-state transducers implement 3x + 1 in base 2 and division by 2 in base 3, and the resulting diagram converts natural numbers from base 3 to base 2. The carries are part of the automaton state. This is the relevant structural lesson here: simultaneous binary and ternary information is natural, but a small pair of valuations discards most of the carry state.

The accelerated map nevertheless gives an exact scalar projection of this binary-ternary interaction. Let n_i be an odd iterate and define

$$k_i = v_2(3n_i + 1), \quad a_i = v_3(n_i + 1), \quad n_{(i+1)} = (3n_i + 1)/2^{(k_i)}.$$

Proposition 3 For every accelerated step,

$$a_{(i+1)} = v_3(3(n_i + 1) + 2^{(k_i)} - 2).$$

More explicitly, the following cases hold.

- If k_i is even, then $a_{i+1} = 0$.
- If $k_i = 1$, then $a_{i+1} = a_i + 1$.
- If $k_i \ge 3$ is odd, put $c(k_i) = 1 + v_3((k_i - 1)/2)$. If $a_i + 1 \ne c(k_i)$, then $a_{i+1} = \min(a_i + 1, c(k_i))$. If the two quantities are equal, cancellation can raise a_{i+1} above their common value.

Proof Multiplying $n_{i+1} + 1$ by $2^{(k_i)}$ gives

$$2^{(k_i)}(n_{(i+1)} + 1) = 3(n_i + 1) + 2^{(k_i)} - 2.$$

Because 2 is a unit in the 3-adic integers, taking v_3 proves the first displayed identity. If k_i is even, then $2^{(k_i)} - 2$ is nonzero modulo 3, so the right-hand side has 3-adic valuation zero. If $k_i = 1$, the correction term vanishes and the valuation rises by one. For odd $k_i \geq 3$, the lifting-the-exponent identity gives

$$v_3(2^{(k_i)} - 2) = 1 + v_3((k_i - 1)/2) = c(k_i).$$

The two summands on the right have valuations $a_i + 1$ and $c(k_i)$. When those valuations differ, the valuation of their sum is the smaller one. When they agree, the next ternary digits determine the amount of cancellation. This proves the case distinction.

The second case gives a precise partial escape theorem. If $k_i = 1$ for t consecutive steps, then

$$n_j + 1 = 3^{j}(n_0 + 1)/2^{j}, \quad a_j = a_0 + j \quad (0 \leq j \leq t).$$

Thus the pair cannot remain in a strip $\{k = 1, a \leq B\}$ for more than $B - a_0 + 1$ consecutive steps. This does not force a large later 2-adic valuation. If the run has exact length t, write $n_0 + 1 = 2^{(t+1)}u$ with u odd. Then $n_t = 2 \cdot 3^{t} u - 1$ and

$$k_t = 1 + v_2(3^{(t+1)}u - 1).$$

For every t, odd choices of u give $k_t = 2$. A large value of $v_3(n_t + 1)$ therefore does not force a compensating 2-adic valuation at the next step.

Failure of pair closure The pair (k_i, a_i) is not a deterministic state for the accelerated map. The starts n = 3 and n = 7 both have (k, a) = (1, 0), but their successors are 5 and 11, with successor pairs (4, 1) and (1, 1). The critical cancellation case also depends on discarded ternary digits: n = 13 and n = 45 both have (k, a) = (3, 0), but their images 5 and 17 have a-values 1 and 2. Consequently, an infinite path in a graph whose vertices are only the pairs (k, a) need not determine an orbit, and classifying paths in that projected graph cannot by itself classify positive Collatz trajectories.

Theorem on unbounded excursions For every integer $L \geq 1$, set $N_L = 2^{(2L+1)} + 1$. For $0 \leq j \leq L$,

$$T^{j}(N_L) = 3^{j} \cdot 2^{(2L + 1 - 2j)} + 1.$$

For each $0 \leq j < L$ the joint state is

$$(k_j, a_j) = (2, 0),$$

while the next state satisfies $(k_L, a_L) = (1, 0)$. Hence runs of the joint state (2, 0) have no universal length bound, even among positive integers, and the valuation immediately after the run need not be large.

Proof The controlled-segment recurrence gives the displayed iterate formula. For $j < L$, the remaining exponent $2L + 1 - 2j$ is at least 3, so $k_j = 2$. At j = 0, the number $N_L + 1 = 2(2^{(2L)} + 1)$ is nonzero modulo 3; for $j \geq 1$, the formula gives $T^{j}(N_L) + 1$ congruent to 2 modulo 3. Thus $a_j = 0$ throughout the segment. At its endpoint,

$$T^{L}(N_L) = 2 \cdot 3^{L} + 1,$$

$$3T^{L}(N_L) + 1 = 2(3^{(L+1)} + 2).$$

The factor in parentheses is odd, so $k_L = 1$; the endpoint plus one is again nonzero modulo 3, so $a_L = 0$.

The Chinese remainder theorem also gives, among odd integers, the relative density

$$\Pr(k = j \text{ and } a = s) = 2^{(1-j)}/3^{(s+1)} \quad (j \geq 1, s \geq 0).$$

Every fixed joint state therefore occurs on a positive-density set. The transition law fails to close on (k_i, a_i): in the equal-valuation case it needs further ternary digits, while k_{i+1} depends on further binary digits of n_{i+1}. Stérin and Woods' full automaton retains those digits and carries, but its base-conversion theorem is a representation theorem rather than a monotone descent theorem. The joint-valuation calculation strengthens the local analysis without supplying universal termination.

Reference: T. Stérin and D. Woods, The Collatz Process Embeds a Base Conversion Algorithm, arXiv:2007.06979, version 4, 2022.

Suffix excess

A useful sufficient condition controls every suffix ending at a selected index r. Suppose that for some $\delta > 0$ and B one has

$$S_r - S_j \geq \delta(r - j) - B \quad (0 \leq j < r).$$

Then the historical burden is bounded by a geometric series:

$$Q_r \leq C_(\delta, B) := 2^B/(2^\delta - 1).$$

The exact threshold in Q_r-form is

$$Q_r < 3n_0 \quad \text{and} \quad S_r > -\log_2(1 - Q_r/(3n_0)).$$

Accordingly, if the displayed constant is less than $3n_0$, the stronger but directly checkable condition

$$S_r > -\log_2(1 - C_(\delta, B)/(3n_0))$$

implies $n_r < n_0$. This condition requires positive backward slope on every suffix, not merely a large value of the total average K_r/r. Proving that every positive orbit supplies such a suitable record time remains open.

Recovery blocks after expanding runs

The suffix condition can be combined with the exact calculation for a long expanding run. Assume that the first ℓ valuations are equal to one. Put $\gamma = \log_2(3/2)$. At the end of this run,

$$M_\ell = (2/3)^\ell = 2^(-\gamma\ell),$$

$$Q_\ell = 3((3/2)^\ell - 1).$$

Let $h_0, \ldots, h_(s-1)$ be the valuations in the following recovery block. For $0 \leq a \leq s$ define

$$H_a = \Sigma_(i = 0)^(a - 1)\ h_i, \quad D_a = H_a - a\log_2 3, \quad D_0 = 0.$$

The quantity D_s is the multiplicative surplus accumulated by the recovery block. Iterating the recurrences for M and Q gives

$$M_(\ell + s) = 2^(D_s - \gamma\ell),$$

$$Q_(\ell + s) = 2^(-D_s)Q_\ell + Q_s^(\text{tail}),$$

$$Q_s^(\text{tail}) = \Sigma_(a = 0)^(s - 1)\ 2^(D_a - D_s).$$

The first term in $Q_(\ell+s)$ is the burden inherited from the expanding run. The second term is generated inside the recovery block. This identity shows why total weighted surplus alone is insufficient: D_s controls the inherited term, whereas the differences $D_s - D_a$ control the new burden.

Recovery block theorem Suppose that for some $\delta > 0$ and B the recovery block satisfies

$$D_s - D_a \geq \delta(s-a) - B \quad (0 \leq a < s).$$

Set

$$C = 2^B/(2^\delta - 1), \quad R = D_s - \gamma\ell.$$

If $C < 3n_0$ and

$$R > \log_2(3(n_0 + 1)/(3n_0 - C)),$$

then $n_{(\ell+s)} < n_0$.

Proof The suffix hypothesis gives $Q_s^{(tail)} \leq C$ by the geometric-series estimate. Since $D_s = R + \gamma\ell$,

$$2^{(-D_s)}Q_\ell = 3 \cdot 2^{(-R)} - 3 \cdot 2^{(-D_s)} < 3 \cdot 2^{(-R)}.$$

Therefore

$$M_{(\ell+s)} = 2^R, \quad Q_{(\ell+s)} < C + 3 \cdot 2^{(-R)}.$$

Substitution into the exact descent certificate gives the strict lower bound

$$M_{(\ell+s)}(1 - Q_{(\ell+s)}/(3n_0)) > 2^R(1 - C/(3n_0)) - 1/n_0.$$

The stated inequality for R is equivalent to

$$2^R(1 - C/(3n_0)) > 1 + 1/n_0.$$

The right-hand side of the descent certificate is consequently greater than one, proving $n_{(\ell+s)} < n_0$.

This theorem combines the two demands that a recovery argument must meet. The total recovery surplus D_s must erase the initial deficit $\gamma\ell$, and every suffix of the recovery block must keep the newly accumulated burden bounded. It does not assert that every positive orbit contains such a block. Establishing that occurrence remains equivalent to supplying the missing global descent argument.

Joint states and exact two phase recovery

The joint-valuation law identifies a particularly transparent recovery pattern. Let ℓ and s be positive integers. Suppose an orbit from $n_0 > 1$ begins with ℓ consecutive valuations equal to 1 and is followed by s consecutive valuations equal to 2. After the first valuation-2 step, the 3-adic coordinate a is zero, and it remains zero throughout the rest of the block. Put

$$\gamma = \log_2(3/2), \quad \delta = \log_2(4/3), \quad R = s\delta - \ell\gamma.$$

Exact two phase criterion For this valuation pattern,

$$M_{(\ell+s)} = 2^R,$$

$$Q_{(\ell+s)} = 3[1 + (3/4)^s((3/2)^\ell - 2)].$$

Consequently, the endpoint lies below n_0 if and only if

$$2^R(n_0 - 1) > n_0 + 1 - 2(2/3)^\ell.$$

Equivalently,

$$R > \log_2((n_0 + 1 - 2(2/3)^\ell)/(n_0 - 1)).$$

Proof The initial expanding run has $M\ell = (2/3)^\ell$ and $Q\ell = 3((3/2)^\ell - 1)$. Each following valuation 2 applies the updates $M \mapsto (4/3)M$ and $Q \mapsto (3/4)(Q + 1)$. Iterating these affine updates gives the displayed formulas. Substitution into $M(1 - Q/(3n_0)) > 1$ simplifies exactly to the stated inequality.

This criterion is sharper than applying the general recovery-block theorem with $\delta = \log_2(4/3)$, $B = 0$, and $C = 3$. The general theorem gives the simpler sufficient condition

$$R > \log_2((n_0 + 1)/(n_0 - 1)).$$

The long (2, 0) excursions from Proposition 3 are therefore favorable, not obstructive: every valuation-2 step contributes the positive surplus δ and supplies the strongest possible linear suffix profile for such a block. The missing global statement is a pointwise theorem forcing a sufficiently long valuation-2 block, or another block with comparable total and suffix surplus, after arbitrary expansion. The joint transition law does not force that occurrence.

Inverse histories and positive density

The inverse-history construction in Mazur's preprint can be translated exactly into the surplus variables above. This translation gives a rigorous extension at the level of positive density, but it does not give pointwise control of every starting value.

Exact dictionary

Fix a forward valuation word $(k_0, \ldots, k_{r-1})$ and list the same valuations in reverse order as $a_t = k_{r-t}$ for $1 \le t \le r$. Put

$$A_t^{(rev)} = \Sigma_{i=1}^{t} a_i, \quad \alpha = \log_2 3.$$

Mazur assigns the complete word the inverse-history weight

$$\omega = 3^r \cdot 2^{-K_r}.$$

Because $S_r = K_r - r\alpha$ and $M_r = 2^{sr}$, this weight is exactly

$$\omega = 2^{-S_r} = 1/M_r.$$

The reverse prefix sums are the forward suffix surpluses:

$$A_t^{(rev)} - t\alpha = S_r - S_{r-t}.$$

Reindexing the exact burden formula therefore gives

$$Q_r = \Sigma_{t=1}^{r} 2^{-(A_t^{(rev)} - t\alpha)}.$$

Thus the preprint's reverse valuation words and the present manuscript's historical burden encode the same finite data. The transfer weight measures reciprocal multiplicative surplus, while Q_r sums the reciprocal surplus of every reverse prefix.

A profile certificate

Profile theorem Let $g(1), \ldots, g(r)$ be real numbers. If

$$A_t^{(rev)} - t\alpha \ge g(t) \quad (1 \le t \le r),$$

and

$$C_g := \Sigma_{t=1}^{r} 2^{-g(t)},$$

then $Q_r \le C_g$. Consequently, the two inequalities

$$C_g < 3n_0,$$

$$S_r > -\log_2(1 - C_g/(3n_0))$$

imply $n_r < n_0$.

Proof The reverse-prefix identity gives 2^(S_(r−t)−S_r) ≤ 2^(−g(t)) for each t. Summing these inequalities gives $Q_r \le C_g$. The bounded-burden certificate then yields

$$M_r(1 - Q_r/(3n_0)) > 1,$$

which is equivalent to $n_r < n_0$. This theorem strictly generalizes the earlier linear suffix condition: choosing g(t) = δt − B recovers the geometric bound.

Reference law and transfer operators

The preprint uses the geometric reference law Pr(a = j) = 2^(−j), whose mean is 2. Its concentration estimate states that for independent reference valuations

$$\Pr(|A_t^{(rev)} - 2t| > v) \le 2\exp(-\min\{v^2/(32t), v/8\}).$$

Fix $0 < \delta < 2 - \alpha$ and put $\lambda = \min\{(2 - \alpha - \delta)^2/32, (2 - \alpha - \delta)/8\}$. Then

$$\Pr(A_t^{(rev)} - t\alpha < \delta t) \le 2e^{(-\lambda t)}.$$

A union bound shows that the probability of a failure at some $t \ge T$ is at most

$$2e^{(-\lambda T)}/(1 - e^{(-\lambda)}).$$

For the finitely many indices $t < T$, the deterministic inequality $A_t^{(rev)} \ge t$ supplies the profile $A_t^{(rev)} - t\alpha \ge \delta t - B$ once $B \ge (\alpha + \delta - 1)T$. Choosing T so that the displayed tail is below one therefore gives positive reference mass to words satisfying a complete suffix-excess profile.

This probabilistic calculation is not yet a statement about a prescribed Collatz orbit. Its value is that Mazur's finite transfer operators are designed to move weighted word restrictions from the reference law to actual integral inverse histories. A profile mask can be inserted into the word family, but the mixing, variation, terminal-selection, and counting estimates must then be rechecked with that mask retained. Such a verification would produce an abundance theorem, not universal coverage.

Source charge and a positive density corollary

Suppose a positive odd source x reaches an odd endpoint R after r accelerated steps with total valuation K_r and positive additive term. The finite-iterate identity gives

$$3^r x < 2^{(K_r)}R.$$

Multiplication by the inverse-history weight therefore yields the source-charge inequality

$$x\omega < R.$$

This is the algebraic reason a lower bound for weighted inverse mass can be converted into a count of distinct integer sources in a fixed size interval.

External positive density corollary Mazur's Theorem 1.1 states that there are explicit constants $c > 0$ and X_0 such that, for every $X \ge X_0$, at least cX positive integers $x < X$ reach 1 in at most (523/50) ln x ordinary Collatz steps. The inverse construction producing these sources uses odd integers. For each such odd $x > 1$, there is an accelerated index r with $n_r = 1$, and hence

$$Q_r < 3x, \quad M_r > 1/(1 - Q_r/(3x)).$$

Section 7 of the preprint gives the stronger accelerated-depth estimate $r \le (34881/10000) \ln x$ for the selected sources. Accordingly, a positive lower-density set has an exact M_r, Q_r descent certificate in

logarithmic accelerated time. This is an immediate corollary of the preprint's stated result; its transfer and mixing argument is not reproduced here.

Limit of the transfer method

The decisive averaging step occurs over finite residue spaces modulo powers of 3. It proves positive surviving mass and selects favorable residue classes or seeds. It does not give a positive lower bound at every residue and does not force a prescribed starting integer into a selected inverse family. An exceptional set of density zero may still contain a particular orbit. A universal proof would require a pointwise covering theorem, an effective argument eliminating every exceptional residue path, or a new forward mechanism that forces a suitable profile time on each orbit. The preprint proves none of these statements.

Reference: L. Mazur, Explicit Positive-Density Collatz Convergence in Logarithmic Time, version 2.1, September 6, 2026, preprint.

Almost everywhere controlled descent

Tao proved that if F(N) tends to infinity, then the minimum of the ordinary Collatz orbit is less than F(N) for logarithmically almost every positive integer N. Equivalently, for any function on the odd positive integers that tends to infinity, the minimum of the corresponding Syracuse orbit is below that function for logarithmically almost every odd starting value. The exceptional set may depend on F, and the theorem does not assert convergence to 1 or give a uniform time at which the small value is attained.

Universal return to a contracting residue

Return lemma Let m be a positive odd integer and write

$$m + 1 = 2^{\ell} u, \quad u \text{ odd}, \quad \ell = v_2(m + 1).$$

For $0 \le j \le \ell - 1$, the accelerated iterates satisfy

$$T^{j}(m) = 3^{j} 2^{(\ell - j)} u - 1.$$

In particular, after $\ell - 1$ accelerated steps the orbit reaches

$$q = 2 \cdot 3^{(\ell - 1)} u - 1 \equiv 1 \pmod 4.$$

Moreover, with $\alpha = \log_2 3$,

$$q + 1 \le (2/3)(m + 1)^{\alpha}.$$

Thus every positive odd orbit reaches the contracting residue class 1 modulo 4 in finitely many accelerated steps. This fact is independent of any density theorem.

Proof The formula is immediate for $j = 0$. If $j \le \ell - 2$, then

$$3T^{j}(m) + 1 = 2(3^{(j+1)} 2^{(\ell - j - 1)} u - 1),$$

and the cofactor is odd. Hence the next valuation is exactly one and induction gives the displayed iterate formula. At $j = \ell - 1$, the number q is one modulo four. Finally, since $\alpha > 1$ and $u \le u^{\alpha}$,

$$q + 1 = (2/3) 3^{\ell} u \le (2/3)(2^{\ell} u)^{\alpha} = (2/3)(m + 1)^{\alpha}.$$

Controlled target theorem

Theorem Let F be any real-valued function on the odd positive integers with F(N) tending to infinity. For logarithmically almost every odd N, the accelerated orbit of N contains an odd integer q such that

$$q < F(N), \quad q \equiv 1 \pmod 4.$$

If q > 1, then T(q) < q. If F(N) < N for all sufficiently large N, the same q is below the original starting value, so the trajectory from N to q satisfies the exact M_r,Q_r descent certificate.

Proof For sufficiently large N define

$$G(N) = ((3/2)(F(N) + 1))^{(1/\alpha)} - 1, \quad \alpha = \log_2 3.$$

Then G(N) tends to infinity. Tao's Syracuse theorem supplies, for logarithmically almost every odd N, an odd orbit value m < G(N). Apply the return lemma to m. The resulting controlled value q lies on the same orbit and satisfies

$$q + 1 \le (2/3)(m + 1)^{\alpha} < (2/3)(G(N) + 1)^{\alpha} = F(N) + 1.$$

Therefore q < F(N). When q > 1, its congruence modulo four gives $v_2(3q + 1) \ge 2$ and T(q) < q. If q < N and q = T^r(N), the exact equivalence from Section 8 gives

$$Q_r < 3N, \quad M_r > 1/(1 - Q_r/(3N)).$$

Logical scope of the combined result

The controlled target theorem strengthens the location of Tao's almost-bounded orbit value: after rescaling the target function, the small value may be chosen in the elementary contracting class 1 modulo 4. It does not force entry into the narrower class 1 modulo 8, and Tao's theorem supplies the existence of the preceding trajectory rather than an explicit formula for its length or valuation vector. Once a finite trajectory is exhibited, the certificate is exactly checkable.

This almost-everywhere statement and Mazur's positive-density convergence statement have different strengths. Tao covers a set of logarithmic density one but proves almost-boundedness rather than convergence. Mazur's stated preprint theorem proves convergence in logarithmic time for a set of positive lower natural density. Neither result eliminates every exceptional starting value, and their conjunction does not prove the Collatz conjecture.

Reference: T. Tao, Almost All Orbits of the Collatz Map Attain Almost Bounded Values, Forum of Mathematics, Pi 10 (2022), e12, doi:10.1017/fmp.2022.8.

Constraint on a hypothetical cycle

If the accelerated odd trajectory forms a cycle of r odd terms, then $n_r = n_0$ and the finite-iterate identity becomes

$$(2^{K_r} - 3^r)n_0 = A_r.$$

Because A_r is positive, every positive nontrivial cycle must satisfy $2^{Kr} > 3^r$. For a fixed valuation vector $(k_0, \ldots, k_{r-1})$, the equation determines at most one candidate value of n_0, namely A_r divided by $2^{Kr} - 3^r$, and a cycle can occur only if this quotient is a positive integer whose successive valuations agree with the proposed vector. This gives a finite exact test for any fixed collection of valuation vectors. It does not exclude cycles of arbitrary length.

9 Comparison of exact and heuristic statements

Table 1 separates the statements established by direct calculation from the interpretations that remain heuristic.

Setting	Valuation	Established statement	Status
$2^x + 1$ with $x \geq 3$	$k = 2$ during the controlled phase	Exact finite recurrence and immediate descent	Exact
$n \equiv 1 \bmod 4$	$k \geq 2$	One accelerated step gives $T(n) < n$	Exact
$n \equiv 3 \bmod 4$	$k = 1$	One accelerated step increases n when $n > 1$	Exact
Odd classes mod 16	Exact or lower bound	Complete one-step refinement	Exact
$2^y m + 1$ with $y \geq 3$	$k = 2$ during the controlled phase	Exact finite recurrence and descent	Exact
Joint 2-adic and 3-adic state	Exact local transition	$k = 1$ raises $v_3(n + 1)$; valuation-2 blocks have an exact recovery criterion	Exact but nonuniform
Sparse expansion	Variable	Later net descent may occur if control persists	Heuristic
Random low bits	Geometric model with mean 2	Negative expected logarithmic drift	Statistical heuristic
Logarithmically almost all odd starts	Orbit-dependent	Reaches $q < F(N)$ with $q \equiv 1 \bmod 4$ for each F tending to infinity	Published theorem plus exact lemma
Selected inverse histories	Variable and transfer-weighted	Positive lower density reaches 1 in logarithmic time	External preprint theorem

Table 1 Exact calculations and heuristic extensions

10 Rigorous reach of modular and binary methods

A residue class modulo a power of 2 determines only as much of the valuation as that modulus can detect. If n is fixed modulo 2^m and $v_2(3n + 1) < m$, the residue determines the valuation exactly. If $3n + 1$ is divisible by 2^m, the same information supplies only the lower bound $v_2(3n + 1) \geq m$. A finer modulus is required to decide whether the valuation is equal to m or larger.

Fixed trailing-bit patterns can give exact information, as Section 6 demonstrates. Closed algebraic forms can also describe individual steps or finite controlled trajectories. These are local results: a sparse starting expansion need not remain sparse, and no fixed modulus determines the complete sequence of valuations along an arbitrary orbit.

The factors $3/2^k$ describe the multiplicative scale of accelerated steps, but iteration also accumulates additive terms arising from the +1 operation. A finite product of the multiplicative factors being smaller than 1 does not by itself prove descent unless the accumulated additive contribution is controlled as well.

A universal descent theorem would state that for every integer n > 1, some later iterate is smaller than n. Such a theorem would prove the Collatz conjecture by strong induction: once an orbit reaches a smaller positive integer, the induction hypothesis applies to that smaller value. Conversely, the Collatz conjecture implies eventual descent because every orbit would reach 1. Uniform eventual descent is therefore equivalent to the conjecture, not a routine consequence of the local calculations.

Status of recent sparse binary claims

A frequently revised manuscript posted on Preprints.org under DOI 10.20944/preprints202401.0227 has discussed binary zero density and Hamming weight. Some versions state descent bounds for sparse subclasses, including logarithmic or quadratic-logarithmic stopping-time bounds, while a later version presents central trajectory assertions as conditional criteria or conjectural behavior. The manuscript is explicitly non-peer-reviewed, and its changing formulations do not provide a stable external theorem that can be imported here.

For that reason, this note does not claim that zero density at least one half implies descent, convergence, or entry into a computationally verified range. Any future use of such a result would require a fixed version, a precise theorem statement, and an independent check of its proof. The exact propositions in this note do not depend on that preprint.

11 Four routes toward a global argument

The following four approaches identify genuine global obstacles. Examining them through the exact iterate formula also reveals two proposed implications that are stronger than the available premises.

Valuation growth

From Section 8, descent at time r is equivalent to

$$K_r > r \log_2 3 + \log_2(1 + A_r/(3\text{^}r\, n_0)).$$

Thus the suggested estimate already has the exact constant C = 1. Proving it for some r on every orbit is not merely sufficient; it is precisely universal eventual descent and hence equivalent to the Collatz conjecture. A larger universal constant C would give a stronger sufficient condition, whereas C < 1 would no longer guarantee descent. The random model predicts typical growth $K_r \approx 2r$, but it supplies no pointwise lower bound for a deterministic orbit.

Contracting residues and long expanding chains

There cannot be a uniform bound on the number of accelerated steps needed after every expanding residue to obtain a net contraction. For any m ≥ 2, start from $n_0 = 2^m - 1$. For $0 \le j \le m - 1$, direct induction gives

$$n_j = 3\text{^}j\, 2\text{^}(m - j) - 1.$$

For $0 \le j \le m - 2$,

$$3n_j + 1 = 2(3\text{^}(j + 1)2\text{^}(m - j - 1) - 1),$$

and the cofactor is odd. Hence $k_j = 1$ and $n_{j+1} > n_j$ for m − 1 consecutive accelerated steps. Since m is arbitrary, expanding chains have unbounded length. Strong induction can still close if each individual start eventually falls below itself, but no fixed recovery horizon can establish that statement.

Cycle vectors

For a fixed complete valuation vector $(k_0, \ldots, k_{r-1})$, the cycle identity determines at most one candidate n_0. Fixing only the length r, however, does not produce a finite check: each k_i is a positive integer with no elementary a priori upper bound, so infinitely many valuation vectors remain. A genuine arbitrary-cycle exclusion therefore needs either a bound on the relevant valuations or totals K_r as a function of r, or a structural argument that handles all such vectors simultaneously. The rational-candidate formula alone supplies neither.

Density estimates

Even an effective estimate showing that the proportion of integers with stopping time greater than N decays faster than every power of 1/N would remain an almost-everywhere statement. A set of density zero can be infinite and can contain any preassigned exceptional integer. Consequently such decay, by itself, cannot prove descent for every concrete starting value. To reach the full conjecture, a density argument must be supplemented by a pointwise mechanism, such as an exhaustive finite residue cover at every height or a computable bound applying to each specified n.

The precise remaining gap

The exact local theory now rules out three tempting shortcuts. The projected pair (k, a) is not a closed state, runs of (2, 0) have unbounded length, and every finite valuation word occurs for infinitely many positive starts. At the same time, long (2, 0) runs are contracting and satisfy an optimal linear suffix profile. The unresolved issue is therefore occurrence, not evaluation: the formulas certify a good block once it appears, but they do not force every orbit to contain one.

Any one of the following pointwise statements would close the gap.

- For every odd $n_0 > 1$, some index r satisfies $Q_r < 3n_0$ and $M_r > 1/(1 - Q_r/(3n_0))$. This is the exact universal descent statement.
- After every expanding prefix, the orbit eventually contains a recovery block whose total surplus exceeds the inherited deficit and whose every suffix satisfies a summable lower profile. The recovery-block theorem would then give descent.
- A refined state retaining enough binary and ternary carry data admits no positive-integer path that avoids all descent certificates. A fixed modulus or the pair (k, a) alone cannot provide such a state.

An average-density statement, an almost-sure 2-adic return theorem, or an enumeration of finite bad patterns does not imply any of these pointwise assertions without an additional argument eliminating the exceptional positive integers. This is the precise place where the present method stops.

12 Anatomy of a hypothetical counterexample

Section 8 gives the exhaustive dichotomy. If an orbit never falls below its starting odd value $n_0 > 1$, it either repeats and forms a nontrivial cycle or it has no repetitions and tends to infinity. The two branches satisfy different exact constraints.

The cyclic branch

For a cycle of r accelerated steps, the valuation vector must satisfy

$$(2^{K_r} - 3^r)n_0 = A_r, \quad 2^{K_r} > 3^r,$$

$$n_0 = A_r/(2^{K_r} - 3^r).$$

The quotient must be a positive integer, and the orbit of that integer must reproduce the proposed valuations. This is a complete finite test for one valuation vector. It does not exclude every cycle because neither the period nor the individual valuations has an a priori bound.

The divergent branch

A divergent counterexample must remain below the critical valuation slope at every scale. The prefix estimate from Section 8 gives, for every r,

$$K_r \le r \log_2(3 + 1/n_0),$$

and the exact logarithmic identity gives the asymptotic restriction

$$\limsup K_r/r \le \log_2 3 \quad \text{as } r \to \infty.$$

This is a necessary condition, not an equality theorem. A divergent orbit may have a smaller liminf or may oscillate; the local formulas do not force its average valuation to converge. The random model's mean value 2 therefore describes typical residue sampling, not a pointwise law that can be imposed on a hypothetical counterexample.

Bounded horizon crossing

Theorem Fix $0 < \delta < 1$ and put

$$R_\delta(n_0) = \operatorname{floor}((1 - \delta)\log_{(3/2)} n_0).$$

There is an effectively computable N_δ such that, for every odd $n_0 \ge N_\delta$ and every $1 \le r \le R_\delta(n_0)$,

$$S_r > 0 \quad \Rightarrow \quad n_r < n_0.$$

Consequently, if the orbit does not descend during this horizon, then every prefix satisfies

$$K_r \le \operatorname{floor}(r \log_2 3) \quad \text{for } 1 \le r \le R_\delta(n_0).$$

Proof Because every valuation is at least one, the recurrence for Q gives the uniform estimate

$$Q_r \le 3((3/2)^r - 1).$$

For $r \le R_\delta(n_0)$, it follows that $Q_r/(3n_0) < n_0^{-\delta}$. If $S_r \ge 1$, the exact certificate $2^{S_r}(1 - Q_r/(3n_0)) > 1$ holds for all sufficiently large n_0. It remains to treat $0 < S_r < 1$.

The nonzero quantity $S_r \log 2 = K_r \log 2 - r \log 3$ is a linear form in logarithms of the algebraic numbers 2 and 3. An effective lower bound for such linear forms gives absolute, computable constants $c > 0$ and $C > 0$ with

$$S_r \ge c\, r^{-C} \quad \text{whenever } 0 < S_r < 1.$$

Here K_r is automatically of order r because $S_r < 1$. Since r is at most a constant multiple of $\log n_0$, the bound $c\, r^{-C}$ eventually exceeds the exponentially smaller quantity $n_0^{-\delta}$. The inequality $1 - 2^{-S} \ge S/2$ for $0 < S \le 1$ then gives

$$Q_r/(3n_0) < 1 - 2^{-S_r},$$

which is equivalent to the exact descent certificate. Thus every positive surplus in the stated horizon causes descent. If no descent occurs, $S_r \le 0$ at every prefix and the integer K_r is at most $\lfloor r$

$\log_2 3$]. The effectivity follows, for example, from Matveev's explicit lower bound for linear forms in logarithms.

Reference: E. M. Matveev, An Explicit Lower Bound for a Homogeneous Rational Linear Form in the Logarithms of Algebraic Numbers II, Izvestiya Mathematics 64 (2000), 1217–1269, doi:10.1070/IM2000v064n06ABEH000314.

Least representatives of noncrossing words

Let w = (k_0, ..., k_{r-1}) be a finite positive valuation word, define K_j and A_r from the finite-iterate identity, and let ρ(w) be the least positive representative of

$$\rho(w) \equiv (2^{K_r} - A_r)3^{-r} \pmod{2^{K_r + 1}}.$$

The inverse 3^{-r} exists modulo $2^{K_r + 1}$. This congruence is equivalent to requiring that the quotient after r prescribed divisions is odd, so its residue class consists exactly of the positive starts that realize w. In particular, every finite word is realized by infinitely many positive integers, while ρ(w) is its smallest one.

For $\alpha = \log_2 3$, define

$$L(r) = \min\{\rho(w) : K_j \le \operatorname{floor}(j\alpha) \text{ for every } 1 \le j \le r\}.$$

This finite minimum is the smallest positive integer capable of following the noncrossing side of the critical line for r prefixes. It converts the unresolved orbit question into an Archimedean growth problem for explicit 2-adic residue representatives.

A sufficient representative growth criterion

Proposition If

$$\liminf_{r \to \infty} L(r)^{1/r} > 3/2,$$

then every sufficiently large odd integer descends below itself. Universal descent would then reduce to checking an effectively bounded finite set of smaller starts.

Proof Let δ satisfy

$$0 < \delta < 2 - \log_2 3$$

$$b = (3/2)^{1/(1-\delta)}.$$

Take δ so small that b is below the displayed liminf. Suppose a sufficiently large n_0 does not descend by $R = R\delta(n_0)$. The bounded-horizon theorem makes its valuation word noncrossing, hence $n_0 \ge L(R)$. Also $K_R \ge R$, and the restriction $\delta < 2 - \log_2 3$ ensures $2^{K_R + 1} > n_0$ for all sufficiently large n_0. The residue class therefore determines n_0 itself as its least representative.

The definition of R gives $n_0 < b^{R+1}$, whereas the liminf hypothesis gives $L(R) \ge \beta^R$ for some $\beta > b$ and all sufficiently large R. The inequalities $n_0 \ge L(R)$ and $n_0 < b^{R+1}$ are then incompatible. This proves descent for all sufficiently large starts.

The proposition is a reduction, not a proof of the required growth. Establishing the strict lower exponential rate for the minimum L(r) is a new global problem; finite-modulus realizability alone gives no such bound.

Common modulus Fourier analysis

A discrepancy bound for one cylinder cannot supply the missing growth. If q = 2^(K_r + 1) and ρ is its least positive representative, the number of members of that cylinder in [1, X] is zero for X < ρ and otherwise equals floor((X − ρ)/q) + 1. Its normalized discrepancy from 1/q is at most 1/X and is exactly zero whenever X is a multiple of q. A positive lower discrepancy bound is therefore impossible, and absolute discrepancy would not in any event imply the one-sided statement ρ > X.

The nontrivial finite object is the union of all noncrossing cylinders at a common modulus. Put

$$\alpha = \log_2 3, \quad m_r = \mathrm{floor}(r\alpha), \quad Q_r = 2^{(m_r + 1)},$$

and let W_r consist of all positive valuation words satisfying $K_j \le \mathrm{floor}(j\alpha)$ for $1 \le j \le r$. Lift every cylinder to modulus Q_r and denote their disjoint union by R_r. The complete cylinder tree is generated by the following multiset recursion:

$$S_0 = \{(0,0)\},$$

$$S_{(r+1)} = \{(K + k, 3A + 2^K) : (K, A) \in S_r, 1 \le k \le m_{(r+1)} - K\}.$$

For a terminal state corresponding to w, its least positive representative is

$$\rho(w) \equiv (2^{(K_r)} - A_r)3^{(-r)} \pmod{2^{(K_r + 1)}}.$$

If c_(r,K) counts the words ending at total K, then

$$c_{(0,0)} = 1, \quad c_{(r+1, L)} = \mathrm{sum}_{(K < L)} c_{(r, K)},$$

$$|R_r| = \mathrm{sum}_K\, c_{(r,K)} 2^{(m_r - K)}, \quad \mu_r = |R_r|/Q_r = (1/2)\mathrm{sum}_K\, c_{(r,K)} 2^{(-K)}.$$

Thus $2\mu_r$ is exactly the conditional Haar probability on the odd 2-adic integers that the geometric valuation walk stays below the critical barrier through time r.

Exact Fourier transform

With e_Q(x) = exp(2πix/Q), define the unnormalized transform by

$$\widehat{1_{R_r}}(t) = \mathrm{sum}_{(a \in R_r)} e_{Q_r}(-ta).$$

A lifted cylinder is a complete arithmetic progression in the common quotient, so its geometric sum vanishes unless its lift count divides the frequency. Hence

$$\widehat{1_{R_r}}(t) = \mathrm{sum}_{(w \in W_r :\ 2^{(m_r - K_r(w))} \mid t)} 2^{(m_r - K_r(w))} e_{Q_r}(-t\rho(w)).$$

Together with the state recursion, this is an exact finite algorithm. Powers of 3 enter through the modular phase ρ(w), especially through 3^{-r} modulo a power of 2; the frequencies themselves remain dyadic. For

$$A_r(X) = \#\{1 \le n \le X : n \bmod Q_r \in R_r\},$$

finite Fourier inversion gives

$$A_r(X) - \mu_r \mathrm{floor}(X) = (1/Q_r) \mathrm{sum}_{(t \ne 0)} \widehat{1_{R_r}}(t) \mathrm{sum}_{(n=1)}^{(\mathrm{floor}\, X)} e_{Q_r}(tn),$$

$$\left|\mathrm{sum}_{(n=1)}^{(\mathrm{floor}\, X)} e_{Q_r}(tn)\right| \le \min(\mathrm{floor}\, X, 1/(2\|t/Q_r\|)).$$

The exact cylinder count is

$$A_r(X) = \mathrm{sum}_{(w \in W_r)} \max(0, \mathrm{floor}((X - \rho(w))/2^{(K_r(w) + 1)}) + 1).$$

Finite spectral test

The companion computation constructs the exact tree, verifies the lifted counts by brute force through eight steps, and evaluates the first 64 nonzero frequencies. Selected results follow. The quantity $v_r\hat{}$ is the transform normalized by $|R_r|$, and the random main term is μ_r ceiling$((3/2)^r)$.

r	Words	μ_r	L(r)	Observed / expected	max $\lvert v_r\tilde{}(t)\rvert$
10	961	0.032257080	27	3 / 1.871	0.027765
12	8,045	0.026062012	27	8 / 3.388	0.009867
14	51,033	0.020123363	27	13 / 5.876	0.005062
16	312,455	0.015458077	27	14 / 10.156	0.001480

Table 2 Common-modulus noncrossing cylinders and low-frequency spectrum

The decreasing low-frequency maximum is evidence of dispersion at these short horizons, not an asymptotic theorem. More importantly, the interval column shows no repulsion from the initial interval: noncrossing starts already occur below $(3/2)^r$, and the orbit of 27 keeps L(r) = 27 through r = 36.

The geometric valuation law provides the relevant scale. Its Cramér rate at β = α is

$$I(\alpha) = \alpha \log(2(\alpha - 1)/\alpha) - \log(\alpha - 1) \approx 0.0549795, \quad \exp(I(\alpha)) \approx 1.05652.$$

The exact count recursion gives $(2\mu_r)^{(-1/r)}$ approximately 1.14164, 1.10820, 1.08721, and 1.07438 at r = 50, 100, 200, and 400. These values move toward the terminal large-deviation scale. Ordinary equidistribution would therefore predict approximately

$$\mu_r(3/2)\hat{}r \approx (1/2)((3/2)/\exp(I(\alpha)))\hat{}r \approx (1/2)(1.4197)\hat{}r$$

survivors below $(3/2)^r$, rather than none. The representative-growth criterion would require the exceptional one-sided avoidance $R_r \cap [1, c\beta^r] = \emptyset$ for some β > 3/2 despite this exponentially growing main term. That is an anti-equidistribution or global rigidity theorem, not a consequence of standard discrepancy decay. The exact recursion and the finite spectrum do not prove such a theorem.

Furstenberg's times-2, times-3 invariant-measure problem is consequently only an analogy. The finite sets R_r change with r and are not restrictions of one probability measure invariant under both commuting maps, so the hypotheses of invariant-measure rigidity are absent.

Why the Sturmian shortcut fails

The maximal mechanical word

$$k_i* = \text{floor}((i + 1)\alpha) - \text{floor}(i\alpha), \quad \alpha = \log_2 3,$$

has prefix sums K_r = floor(rα) and begins 1, 2, 1, 2, 1, 2, 2, 1, 2, 1, 2, 2, ... It has no consecutive 1s, and its runs of 2s have length one or two. These facts are the reverse of the often suggested claim that the same word has isolated 2s. More importantly, the inequalities $K_j \le$ floor(jα) do not force equality. Many other noncrossing words exist, so this mechanical word does not determine L(r).

The orbit of 27 is a concrete warning. Its valuations begin

$$1, 2, 1, 1, 1, 1, 2, 2, 1, 2, 1, 1, 2, 1, 1, 1, 2, 3, \ldots .$$

They satisfy $K_j \le \mathrm{floor}(j\alpha)$ through $j = 36$. At $j = 37$ the total jumps to $K_{37} = 59 > \mathrm{floor}(37\alpha) = 58$, and the iterate is $23 < 27$. Hence $L(r) \le 27$ for every $r \le 36$. By contrast, direct evaluation of the maximal mechanical word at $r = 36$ gives $\rho(w^*) = 72{,}410{,}175{,}582{,}919{,}675$. The least-representative problem must control all admissible words, not one balanced word.

Adelic exclusion of rational weighted words

The remaining divergent branch can be narrowed further without assuming periodicity. Let an infinite valuation word be realized by a fixed positive odd integer n_0, put $h_j = K_j - j$, and define its weighted generating function

$$G_w(z) = \sum_{j=0}^{\infty} 2^{h_j} z^j.$$

The finite-iterate identity gives, for every r,

$$n_0 = 2^{K_r} n_r / 3^r - \sum_{j=0}^{r-1} 2^{K_j} / 3^{j+1}.$$

In the 2-adic numbers the first term on the right tends to zero, because n_r and 3 are odd and $K_r \ge r$. Therefore every infinite positive orbit satisfies the exact 2-adic identity

$$\sum_{j=0}^{\infty} 2^{K_j} / 3^{j+1} = -n_0 \quad \text{in } \mathbb{Q}_2.$$

Since $2^{K_j} = 2^j 2^{h_j}$, the same series can be written as

$$\sum_{j=0}^{\infty} 2^{K_j} / 3^{j+1} = (1/3) G_w(2/3).$$

Call the word permanently subcritical if there are $\varepsilon > 0$ and j_0 such that $K_j \le (\log_2 3 - \varepsilon) j$ for every $j \ge j_0$. Under this hypothesis the displayed series also converges in the real numbers, where every term is positive.

Theorem No permanently subcritical valuation word of a positive integer has $G_w(z)$ rational over $\mathbb{Q}$.

Proof Assume $G_w(z)$ belongs to $\mathbb{Q}(z)$. Its Taylor series converges at $z = 2/3$ in both the real and 2-adic absolute values, so both sums equal the same rational evaluation $G_w(2/3)$. The real value is strictly positive, whereas the 2-adic identity requires $G_w(2/3) = -3n_0 < 0$ as a rational number. This is impossible.

Every eventually periodic subcritical valuation word is covered. Indeed, if an eventual period has length p and total valuation H, then $h_{j+p} - h_j = H - p$ on the periodic tail, so the tail of G_w is a finite sum of geometric series and G_w is rational. The theorem is stronger: it excludes any subcritical word whose weighted prefix sequence 2^{h_j} has a rational ordinary generating function. It does not exclude an arbitrary aperiodic word.

Band growth and sparse aperiodic excess

A second exact estimate excludes a different aperiodic regime. Put

$$\alpha = \log_2 3, \quad c = \alpha - 1 = \log_2(3/2), \quad h_r = K_r - r.$$

Theorem If a positive odd integer n_0 realizes a valuation word of length r, and m is the number of indices $i < r$ for which $k_i \ge 2$, then

$$\log_2(n_0 + 1) \ge c(r+1)/(\alpha^{m+1} - 1) \ge c(r+1)/(\alpha^{h_r+1} - 1).$$

Proof Split the word at its m reset indices $k_i \ge 2$. This produces $m + 1$, possibly empty, runs of $k = 1$. Let ℓ_j be the length of the jth run and t_j its starting index. During a $k = 1$ step, $x_i = n_i + 1$ obeys $x_{i+1} = 3x_i/2$. A

run of length ℓ_j therefore requires 2^(ℓ_j+1) to divide x_(t_j); the same inequality is trivial when $\ell_j = 0$. On the other hand, every accelerated step satisfies $x_{i+1} \le (3/2)x_i$. With $L = \log_2(n_0 + 1)$, this gives

$$\ell_j + 1 \le \log_2 x_(t_j) \le L + ct_j.$$

For $j < m$, the next run starts at $t_{j+1} = t_j + \ell_j + 1$, hence $t_{j+1} \le \alpha t_j + L$. Induction yields

$$t_m \le L(\alpha\^m - 1)/(\alpha - 1).$$

The final run gives $r + 1 = t_m + \ell_m + 1 \le \alpha t_m + L$. Substitution proves the first inequality. Each reset contributes at least one unit to $h_r = \Sigma(k_i - 1)$, so $m \le h_r$; monotonicity of the denominator proves the second inequality.

Applied to the least representative ρ(w), the theorem gives

$$\rho(w) + 1 \ge 2\^(c(r + 1)/(\alpha\^(h_r + 1) - 1)).$$

If one fixed n_0 realizes all prefixes of an infinite word, then necessarily

$$h_r \ge \log_\alpha(1 + c(r + 1)/\log_2(n_0 + 1)) - 1.$$

Consequently no positive integer can realize an infinite valuation word with $h_r = o(\log r)$. More quantitatively, if $h_r + 1 \le (1 - \varepsilon)\log_\alpha(r + 1)$ along an unbounded sequence of indices, then

$$\rho(w) + 1 \ge 2\^(c(r + 1)\^\varepsilon)$$

along those indices, so the corresponding least representatives cannot remain equal to one fixed starting integer.

The surviving aperiodic corridor

These two exclusions are complementary but do not finish the divergent case. A permanently subcritical counterexample would now have to be genuinely aperiodic and satisfy, for some $\varepsilon > 0$,

$$\log_\alpha r + O(1) \le h_r \le (c - \varepsilon)r,$$

$$G_w(z) \notin Q(z),$$

$$(1/3)G_w(2/3) = -n_0 \quad \text{in } Q_2,$$

even though the same convergent series has a positive real value. The different real and 2-adic limits are not contradictory for a general irrational series. Excluding this corridor would require a new rigidity theorem tying Collatz admissibility to the arithmetic nature of G_w, or a stronger lower bound on the least representatives. Thus the aperiodic obstruction has been narrowed, not removed.

Kernel regularity does not follow from admissibility

Fix an integer base $d \ge 2$ and write $a_j = 2\^(h_j)$. The d-kernel of the coefficient sequence $a = (a_j)_{j \ge 0}$ is

$$K_d(a) = \{(a_(d\^e n + r))_(n \ge 0) : e \ge 0, 0 \le r < d\^e\}.$$

Over Q, the sequence is d-regular when the Q-vector space spanned by this kernel is finite-dimensional. Finite generation of the corresponding Z-module is the usual integral form and is stronger. This is the precise property needed before automatic-sequence or Mahler-function machinery can be inferred from the coefficient sequence itself. The finite-state character of a base-conversion device does not, by itself, establish this index-regularity.

A positive-orbit obstruction

Proposition Collatz admissibility does not imply a finite-dimensional d-kernel in any base $d \geq 2$.

Proof The accelerated fixed point $n_j = 1$ has $k_j = 2$, $K_j = 2j$, $h_j = j$, and $a_j = 2^j$. Its d-kernel contains the subsequences

$$b_e(n) = a_(d^e n) = (2^(d^e))^n, \quad e = 0, 1, 2, \ldots .$$

For any finite selection of distinct indices e, the corresponding exponential sequences are Q-linearly independent: evaluating them at consecutive n gives a Vandermonde matrix with distinct bases $2^{(d^e)}$. Thus the kernel span is infinite-dimensional. The same obstruction holds for every positive orbit that eventually reaches 1, because its coefficients are eventually $C2^j$ for a fixed nonzero rational C. Altering finitely many initial terms cannot restore regularity, since finite-support sequences are d-regular. Hence d-regularity is not even a general property of known convergent positive orbits.

What local 2-adic admissibility permits

The coefficient recurrence is

$$a_(j + 1) = 2^(k_j - 1)a_j.$$

Conversely, let (h_j) be any nondecreasing sequence of nonnegative integers with $h_0 = 0$ and set

$$k_j = 1 + h_(j + 1) - h_j.$$

Then every k_j is positive. Each finite valuation prefix is realized by one odd residue class modulo $2^{(K_r + 1)}$, and the compatible nested classes determine a unique odd 2-adic starting value. Therefore local Collatz admissibility allows essentially arbitrary nondecreasing exponent sequences h_j. The extra requirement that the same prefixes arise from one ordinary positive integer is Archimedean: the least positive representatives must eventually stabilize at that integer. No finite-kernel conclusion follows from the local residue construction alone.

Aperiodic regular and nonregular models

Nonregular model Choose an irrational number β with $0 < \beta < \log_2(3/2)$, and put

$$h_j = \mathrm{floor}(\beta j), \quad k_j = 1 + \mathrm{floor}(\beta(j + 1)) - \mathrm{floor}(\beta j).$$

This is an aperiodic word with k_j in {1, 2} and K_j/j tending to $1 + \beta < \log_2 3$, so it remains strictly below the critical valuation slope. Yet $a_j = 2^{(\mathrm{floor}(\beta j))}$ grows exponentially. Every sequence with finite-dimensional d-kernel over Q has at most polynomial growth: a finite linear representation reads the base-d digits of j using O(log j) matrices of fixed norm. Consequently this aperiodic admissible 2-adic model is not d-regular for any base d.

Regular model For integers $d \geq 2$ and $m \geq 1$, put

$$h_j = m\,\mathrm{floor}(\log_d(j + 1)), \quad a_j = 2^{(m\,\mathrm{floor}(\log_d(j + 1)))}.$$

The resulting valuation word is aperiodic and locally admissible, while $h_j = O(\log j)$ keeps it permanently subcritical. Its block recursion makes a d-regular, and its generating function $A(z) = \Sigma_{j \geq 0} a_j z^j$ satisfies the Mahler equation

$$A(z) = 1 + z + \cdots + z^{(d - 2)} + 2^m z^{(d - 1)}(1 + z + \cdots + z^{(d - 1)})A(z^d).$$

Taking $m \geq \mathrm{ceiling}(\log_\alpha d)$, where $\alpha = \log_2 3$, gives the same logarithmic order required by the band-growth lower bound. This provides a structured Mahler test case, not a positive-integer orbit: its nested residue representatives still have to satisfy the global stabilization condition.

Table 3 records an exact computation for the regular test case d = 2 and m = 2. The bit length of the least positive representative ρ_r nearly fills the available modulus of $K_r + 1$ bits. This supports nonstabilization in this example but is computational evidence only.

Horizon r	Total K_r	Bit length of ρ_r
8	14	15
16	24	25
32	42	43
64	76	76
128	142	142
256	272	273

Table 3 Least representatives for the regular d = 2 and m = 2 model

The logarithmic regular slice

Suppose a hypothetical infinite positive orbit had a d-regular coefficient sequence. Polynomial growth would give $h_j \le C \log j$ for some constant C, whereas the band-growth theorem gives $h_j \ge \log_\alpha j + O(1)$. Such an orbit would therefore be confined to the narrow strip

$$\log_\alpha j + O(1) \le h_j \le C \log j.$$

In particular, any corridor with $h_j/\log j$ tending to infinity, including linear excess, is automatically nonregular. Mahler theory can therefore address only a selected logarithmic subclass, and the regular model above shows that this subclass is not empty at the 2-adic level. What remains missing is a theorem that ordinary positive-integer realizability forces regularity, algebraicity, or some other comparable rigidity; present Collatz admissibility supplies none of them.

Reference: J.-P. Allouche and J. Shallit, The Ring of k-Regular Sequences, Theoretical Computer Science 98 (1992), 163–197, doi:10.1016/0304-3975(92)90001-V.

The ternary coordinate does not close the argument

Along the mechanical word, Proposition 3 resets $a = v_3(n + 1)$ to zero at each k = 2 step. Because no two k = 1 steps are consecutive, a is at most one after the first reset. This bounded ternary valuation is consistent with the global identity

$$A_r + 2^{K_r} = 2^{K_r}(n_r + 1) - 3^r n_0.$$

Indeed, if $a_r < r + v_3(n_0)$, then the two terms on the right have different 3-adic valuations and

$$v_3(A_r + 2^{K_r}) = a_r.$$

A bound $a_r \le B$ therefore does not imply that $A_r + 2^{K_r}$ is divisible by 3^{r-B}; it normally implies the opposite. The additive term supplies exactly the carry information needed to keep the 2-adic and 3-adic descriptions consistent. No contradiction follows from tracking these two scalar valuations alone.

The remaining target is now explicit. A proof must either establish sufficient growth of L(r), force a surplus crossing by another pointwise invariant, or exclude all cycle candidates. The bounded-horizon

theorem narrows where a counterexample can hide, but it does not show that the exceptional residue path is absent.

13 Scale invariance finite 2 adic shadows and corridor escape

The exact finite-word law, the least-representative formulation, and large-bit computations explain three initially surprising observations. Uniformly sampled odd integers behave almost identically at every large bit scale. Specially selected integers can avoid the critical line for thousands of accelerated steps. Once the critical line is crossed, the same step is normally already a strict descent. The first and third observations have rigorous finite-horizon explanations. The second is explained by finite shadowing of exceptional 2-adic trajectories. What remains unproved is that every such shadow must escape when its starting point is one fixed ordinary positive integer.

Exponential probabilistic repulsion

For a Haar-random odd 2-adic integer, the accelerated valuations are independent and have the exact geometric law

$$\Pr(k_i = m) = 2^{(-m)}, \quad m \geq 1.$$

Hence $E(k_i) = 2 > \alpha = \log_2 3$, and the surplus $S_r = K_r - r\alpha$ has positive mean increment

$$E(S_{(r+1)} - S_r) = 2 - \alpha \approx 0.41504.$$

Remaining in the noncrossing corridor through time r requires $K_j \leq j\alpha$ at every prefix $1 \leq j \leq r$. This event is contained in the terminal lower-tail event $K_r \leq r\alpha$. The Chernoff-Cramér estimate from Section 12 therefore gives

$$\Pr(K_j \leq j\alpha \text{ for all } 1 \leq j \leq r) \leq \exp(-I(\alpha)r),$$

$$I(\alpha) = (\alpha - 1)\log(2(\alpha - 1)/\alpha) + \log(2/\alpha) \approx 0.0549795.$$

The exponential base of this upper bound is $\exp(-I(\alpha)) \approx 0.946505$. The all-prefix restriction makes the exact probability smaller than the terminal-event bound. Exact recursion gives survival probabilities $3.4515487165 \times 10^{-5}$, $5.4567042148 \times 10^{-8}$, $3.4368182949 \times 10^{-13}$, and $5.8038516047 \times 10^{-19}$ at r = 100, 200, 400, and 630, respectively. The strong law consequently implies that an infinite noncrossing word has Haar measure zero. This is an almost-everywhere obstruction, not a pointwise obstruction: a measure-zero set may still contain an exceptional ordinary positive integer.

Exact counts and samples from 100 to 1000 bits

For an odd integer selected uniformly from $[2^{B-1}, 2^B)$, the leading bit is fixed and the lower B − 1 bits are uniform. Every noncrossing valuation cylinder is therefore counted exactly while $\text{floor}(r\alpha) + 1 \leq B - 1$. At the largest horizon satisfying this condition, the exact data are shown in Table 4.

Bits B	Exact horizon	Probability	Exact surviving starts
100	62	$5.2792767878 \times 10^{-4}$	$1.673069597 \times 10^{26}$
200	125	$6.6374883512 \times 10^{-6}$	$2.666508137 \times 10^{54}$
500	314	$5.5732621751 \times 10^{-11}$	$4.560866014 \times 10^{139}$
1000	630	$5.8038516047 \times 10^{-19}$	$1.554719237 \times 10^{282}$

Table 4 Exact noncrossing counts at the full bit-compatible horizon

The rows refer to different horizons. At any fixed common horizon, the probability is independent of B because only the required low binary residues are involved. Increasing the bit length extends the horizon for which the residue calculation is exact. It also shows why tiny density is not the same as emptiness: an extremely small proportion can still represent enormously many starting values.

Two million uniformly selected odd starts were computed at each bit size with exact arbitrary-precision arithmetic. The observed first-crossing quantiles are listed in Table 5.

Bits	Median	90%	99%	99.9%	99.99%	Latest
100	1	8	27	54	86	191
200	2	8	27	54	85	157
500	1	8	27	54	85	171
1000	1	8	27	54	85	231

Table 5 First critical-line crossing times in two million samples per scale

All eight million sampled starts crossed and descended below their starting value. In every sample, the first descent occurred at the first critical-line crossing, and no sample remained unresolved after 2000 accelerated steps. These computations illustrate the exact scale invariance and probabilistic drift; they cannot exclude a density-zero exceptional orbit.

Engineered starts as finite 2 adic shadows

The family $2^B - 1$ has a precise 2-adic interpretation:

$$2\text{^}B - 1 \equiv -1 \pmod{2\text{^}B}.$$

The negative integer −1 is a fixed point of the accelerated map and has valuation word (1, 1, 1, ...). Its positive truncation shadows this word exactly:

$$T\text{^}j(2\text{^}B - 1) = 3\text{^}j\, 2\text{^}(B - j) - 1, \quad 0 \le j \le B - 1.$$

The first B − 1 valuations are therefore all equal to 1. As B increases, these values are not longer prefixes of one fixed positive integer. Instead, $2^B - 1$ tends to −1 in the 2-adic integers while tending to positive infinity in the real numbers. Exact calculations are summarized in Table 6.

B	First crossing and descent	K at crossing	Final k	Peak bits
100	229	366	7	158
200	469	744	3	317
500	1153	1828	6	792
1000	2250	3567	4	1585

Table 6 Exact behavior of the forced expanding family 2^B minus 1

An exhaustive calculation of all odd offsets $|d| \le 10001$ around each 2^B found that d = −1 had the latest descent in every range. The maximum transient bit length is approximately $B \log_2 3$, as predicted by the forced initial expansion.

The negative accelerated cycle −5 → −7 → −5 gives a second example. Its periodic valuations are (1, 2, 1, 2, ...), of average 3/2 < α. The positive integers $2^B - 5$ shadow this cycle until their finite upper boundary intervenes. Their first crossings and descents occur at steps 69, 168, 434, and 831 for B = 100, 200, 500, and 1000.

For $2^B - 1$, the initial valuation-one block creates a deficit of about $(\alpha - 1)B \approx 0.58496B$ relative to the critical line. Once the negative fixed point is no longer shadowed, a typical valuation sequence recovers surplus at mean rate 2 − α. This gives the heuristic recovery scale

$$B + ((\alpha - 1)/(2 - \alpha))B \approx 2.409B,$$

consistent with the observed linear crossing times. This recovery estimate is explanatory rather than a pointwise theorem.

Why a first crossing normally gives descent

Put $S_r = K_r - r\alpha$ and define the weighted prefix history

$$P_r = \text{sum_}(j = 0)\text{^}(r - 1)\ 2\text{^}(S_j).$$

The exact finite-iterate identity becomes

$$n_r = 2\text{^}(-S_r)(n_0 + P_r/3).$$

Consequently, $n_r < n_0$ if and only if

$$2\text{^}(S_r) > 1 + P_r/(3n_0).$$

If r is the first critical-line crossing, then $S_j \le 0$ for every $j < r$, and hence $P_r \le r$. The explicit condition

$$2\text{^}(S_r) > 1 + r/(3n_0)$$

therefore guarantees descent at that crossing. Since the least possible crossing total is ceiling(r $\log_2 3$), every $n_0 \ge 2^{B-1}$ descends at a first crossing at time r whenever

$$3 \cdot 2\text{^}(B - 1)(2\text{^}(\text{ceiling}(r \log_2 3)) - 3\text{^}r) > r3\text{^}r.$$

Direct integer verification for every $1 \le r \le 10000$ gives the safety factors in Table 7.

Bits B	Minimum $\log_{10}$ safety factor through 10000 steps
100	22.9101
200	53.0131
500	143.3221
1000	293.8371

Table 7 Exact first-crossing descent margins

The additive burden is therefore negligible over this checked horizon. The statement is deliberately finite. Exceptionally close rational approximations to $\log_2 3$ prevent this computation from being promoted to an unproved assertion uniform over every r.

The exact representative escape obstruction

For each horizon define the nested survivor set

$$C_r = \{n > 1 \text{ odd} : K_j(n) \le \operatorname{floor}(j\alpha) \text{ for } 1 \le j \le r\},$$

$$L(r) = \min C_r.$$

Proposition No positive integer has an infinite noncrossing orbit if and only if L(r) tends to infinity.

Proof If one positive integer n_0 survives every horizon, then $L(r) \le n_0$ for every r. Conversely, the sets C_r are nested, so their integer minima are nondecreasing. If L(r) is bounded, it eventually stabilizes at an integer belonging to every C_r, which produces an infinite noncrossing orbit.

Equivalently, along every nested infinite noncrossing valuation word, the least positive cylinder representatives must tend to infinity rather than eventually stabilize at one positive integer. The families $2^B - 1$ and $2^B - 5$ exhibit precisely this escape: their shadowing times increase only because their positive representatives increase without bound, while their 2-adic limits are negative periodic points.

The exact probability estimates prove that the density of C_r tends to zero exponentially. They do not prove that its least element tends to infinity. Every finite experiment is compatible both with universal representative escape and with one density-zero aperiodic corridor stabilizing at an ordinary positive integer. Excluding the latter possibility is the global rigidity theorem still missing from the divergent branch of the Collatz problem.

14 Origin sensitive computation and mixed residue diagnostics

Three concrete extensions were pursued: a larger exact search for the least noncrossing representative, an operator norm that detects the Archimedean origin, and an exact comparison of the binary starting cylinder with the dynamically forced ternary tail class. A fourth calculation quantifies the finite safe-crossing window supplied by close powers of two and three. The search gives new certified finite data. The operator and mixed-residue calculations sharpen the obstruction rather than close it.

Certified least representatives below one hundred billion

A parallel exact scanner tested every odd starting value below 10^{11}. It computes the barrier $\operatorname{floor}(r \log_2 3)$ from the bit length of 3^r and switches from a 128-bit fast path to arbitrary-precision limb arithmetic

before any possible overflow. An independent replay retains the complete valuation word of each global record. The new records are listed in Table 8.

Horizons r	Exact L(r)	First crossing	K at crossing
249–250	1,200,991,791	251	398
251–272	1,827,397,567	273	437
273–281	2,788,008,987	282	447
282–344	12,235,060,455	345	547

Table 8 New certified least noncrossing representatives

$$L(345) > 99{,}999{,}999{,}999.$$

The last record word has length 345 and total valuation 547. Its valuation multiplicities are 219 occurrences of 1, 77 of 2, 31 of 3, 12 of 4, and three each of 5 and 6. Its mean valuation 547/345 = 1.585507... lies only slightly above $\log_2 3$ at its first crossing. Over the finite certified range, the smallest value of log L(r)/r is 0.0675220... at r = 344, still above the Haar Chernoff exponent 0.0549794.... This comparison is finite evidence only. A later plateau of L(r) could change it.

The exact anchored operator

Let σ ≥ 0 be the current noncrossing budget. If the next valuation is k and the next odd value is m, the inverse branch is

$$n = (2^k m - 1)/3, \quad m \equiv (-1)^k \pmod 3, \quad \sigma' = \sigma + \alpha - k.$$

Thus an exact recursion for anchored counts must retain both the budget and a selected residue modulo 3; after t inverse steps it retains a residue modulo 3^t. For a nonnegative function f define its anchored average at X by

$$A_f(X) = X^{-1} \sum_{1 \le n \le X} f(n).$$

The anchored weak norm $\|f\|_A$ is the least upper bound of A_f(X) over integers X ≥ 1. A permitted point mass at m is mapped to a point mass at n. The exact norm ratio is

$$\|\delta_n\|_A / \|\delta_m\|_A = m/n = 3m/(2^k m - 1) > 3/2^k.$$

The k = 1 inverse branch therefore expands this norm. A positive one-step operator cannot inherit the Haar contraction factor without a multi-step arithmetic dispersion theorem.

For a complete period $Q_r = 2^{\lfloor r\alpha \rfloor + 1}$, let $a_1 < \cdots < a_M$ be the surviving residues. Periodicity gives the exact formula

$$\|1_{(R_r)}\|_A = \max\{M/Q_r, \max_j j/a_j\}.$$

r	Haar density	Anchored supremum	Amplification
6	0.0625000	0.0745098	1.192
10	0.0322571	8/111 = 0.0720721	2.234
14	0.0201234	8/111 = 0.0720721	3.582
18	0.0118942	8/111 = 0.0720721	6.059

Table 9 Exact origin-sensitive maxima over one complete period

The average density falls while eight small survivors below 111 hold the anchored maximum fixed. This is the concrete one-atom obstruction hidden by Haar averages and low-frequency spectral estimates.

Binary and ternary residue coupling

For a valuation prefix p of length t, define $A_0 = 0$ and $A_{i+1} = 3A_i + 2^{K_i}$. The finite-iterate identity

$$2^{K_t} n_t = 3^t n_0 + A_t$$

forces the current tail into the exact ternary class

$$n_t \equiv c_p := A_t 2^{-K_t} \pmod{3^t}.$$

Conversely, a positive odd tail m in this class with $2^{K_t} m > A_t$ reconstructs the start as

$$n_0 = (2^{K_t} m - A_t)/3^t.$$

The budgeted-tail recursion is therefore a mixed minimum in one dynamically selected ternary class, not an intersection with an independent random class. Exact enumeration of all 663,535 noncrossing words of length 17 makes the distinction visible. The ternary marginal has star discrepancy 0.001304, but the normalized binary and ternary coordinates have Pearson correlation 0.49998. There are 6,731 binary and 6,686 ternary residues in their respective bottom one percent, with 3,426 simultaneous occurrences. Independence would predict only 67.8. A small marginal discrepancy therefore does not imply the joint anti-clustering required at the origin.

Exact Diophantine safe crossing windows

At a first critical-line crossing, $P_r \le r$. With $m_r = \lceil r \log_2 3 \rceil$, descent is forced whenever

$$n_0(2^{m_r} - 3^r) > r3^{r-1}.$$

The exact least sufficient starting size at that horizon is

$$B(r) = \left\lfloor r3^{r-1}/(2^{m_r} - 3^r) \right\rfloor + 1.$$

r	m_r	Record B(r)
15,601	24,727	285,814,987
47,468	75,235	1,447,674,322
79,335	125,743	7,216,089,271
190,537	301,994	984,572,779,225

Table 10 Record finite first-crossing thresholds through step 200000

Consequently, if $n_0 \ge 984{,}572{,}779{,}225$ and its first critical-line crossing occurs by step 200,000, that crossing already gives $n_r < n_0$. The record spikes align with upper and intermediate convergents of $\log_2 3$. This is a large finite safe window, not a proof that a crossing occurs within it.

Together these calculations sharpen the remaining target. A viable operator theorem must prove multi-step anti-clustering for the exact joint binary and ternary state at its distinguished positive lift. A scalar spectral gap, a small ternary marginal discrepancy, or a stronger irrationality measure for $\log_2 3$ does not by itself control L(r).

15 Lean verification

The companion file CollatzLocal.lean formalizes the exact arithmetic core in Lean 4 with Mathlib. The file was compiled with no errors. Its checked theorems establish the following statements.

- Numbers of the form $4q + 1$ have an odd-step numerator divisible by 4, and division by 4 produces a smaller value when q is positive.
- Numbers of the form $4q + 3$ have an odd-step numerator equal to $2(6q + 5)$, where the cofactor is odd, and the resulting value is larger.
- The trailing-ones expression has exactly one factor of 2 after the odd step because its displayed cofactor is odd.
- The resulting trailing-ones accelerated value is strictly larger than the starting value whenever the latter exceeds 1.
- For $x \geq 3$, written as $x = y + 3$, the value $2^x + 1$ has the stated factorization by 4, an odd cofactor, and a strictly smaller first accelerated value.
- The residue classes $8q + 1$ and $8q + 5$ have the stated factorizations, valuation information, and one-step decreases.
- All odd residue classes modulo 16 have the stated exact valuations or lower bound, with the corresponding one-step behavior.
- The families $2^x + 3$ and $2^x + 2^y + 1$ in the stated ranges have the asserted first-step factorizations and inequalities.
- Every value $2^y m + 1$ with $y \geq 3$ has the general factorization by four and a smaller first accelerated value.
- The example $T(9) = 7$ verifies that multiplication by 3 creates a two-bit block, so binary support does not undergo a simple uniform shift.
- Two prescribed accelerated steps satisfy $2^{k+l}c = 9a + 3 + 2^k$, explicitly exhibiting the additive correction.
- The family $3^a 2^{b+1} - 1$ has the displayed one-step factorization underlying arbitrarily long expanding chains.
- The normalized real-variable certificate is equivalent to descent, and the P/M form gives the weighted-history ratio used in the surplus-record formulation.
- At a first critical-line crossing, termwise prefix weights bounded by one give $P_r \leq r$; the explicit threshold $M_r > 1 + r/(3n_0)$ then yields strict descent.
- A geometric recurrence bounds the weighted history under persistent failure of the certificate.
- A geometric suffix estimate and the corrected threshold $1/(1 - C/(3n))$ give a rigorous sufficient descent condition once the burden Q is bounded by $C < 3n$.
- The recovery-block certificate converts the combined inherited and tail burden bound $Q < C + 3/X$ into descent once X exceeds $3(n + 1)/(3n - C)$.
- The inverse-history weight is the reciprocal surplus $1/M$, reverse valuation prefixes are forward suffix surpluses, and a general profile bound yields a finite descent certificate.
- The source-charge inequality converts weighted inverse mass into an integer count; the cited positive-density theorem is used only as an external corollary, not as a universal statement.

- For any nested nonempty family of subsets of the natural numbers with specified least representatives L(r), absence of a common survivor is equivalent to L(r) tending to infinity. This checks the order-theoretic reduction but supplies no Collatz-specific growth estimate.
- The finite-iterate identity forces the exact ternary tail residue after reduction modulo 3^t.
- Every inverse branch satisfies the strict cross-multiplied scale inequality $3n < 2^k m$, and a valuation-one branch strictly increases the odd value.
- The power-gap inequality $r3^r < 3n(2^m - 3^r)$ implies the normalized first-crossing threshold used in the finite Diophantine window.

The formalization intentionally stops at these local theorems. It contains no theorem asserting termination, universal descent, the absence of nontrivial cycles, or the Collatz conjecture. The bounded-horizon theorem, the exhaustive search below 10^{11}, any Collatz-specific asymptotic lower bound for L(r), the common-modulus Fourier analysis, the adelic generating-function theorem, the band-growth estimate, and the d-kernel results in Section 12 are not formalized in the companion file. Successful compilation confirms that the encoded finite arguments are accepted by Lean; it does not enlarge their mathematical scope.

16 Conclusion

Every odd $O > 1$ congruent to 1 modulo 8 admits an exact finite decreasing segment. If e_1 is its lowest positive binary exponent, the segment has $J = \lfloor (e_1 - 1)/2 \rfloor$ accelerated steps and is independent of the higher gaps and the Hamming weight; the family $2^x + 1$ is a special case. Residues modulo 4 also give a complete one-step classification: values congruent to 1 decrease, while values congruent to 3 increase. Beyond these local facts, sparse binary structure and random valuation models provide descriptive heuristics rather than uniform control.

The elementary inequality remains valid and useful inside algebraic expressions that have been derived from an actual orbit. It does not justify evolving binary components separately, and it does not determine future valuations. Stérin and Woods' base-conversion automaton explains why binary carries and ternary division belong in one state space. The exact joint-valuation law extracted here controls runs with $k = 1$, proves that the projected pair is not a closed state, and exhibits arbitrarily long $(k, a) = (2, 0)$ segments. Those segments are contracting recovery blocks; the unresolved problem is to force a sufficiently strong recovery profile after arbitrary expansion. Tao's almost-boundedness theorem, combined with the elementary return lemma, shows that for every divergent target F, logarithmically almost every odd orbit reaches a value $q < F(N)$ congruent to 1 modulo 4 and hence obtains an immediate local descent certificate. The exceptional set may depend on F, and the theorem does not give an explicit witness time. Mazur's inverse-history method adds a rigorous positive-density logarithmic-time result and identifies the inverse weight with reciprocal surplus, but its residue averaging does not cover every starting value. The bounded-horizon theorem further shows that a sufficiently large non-descending start must keep every valuation prefix below the critical line through a logarithmic horizon. The least-representative function L(r) packages the missing Archimedean obstruction: growth faster than $(3/2)^r$ would force descent outside an effective finite range. The common-modulus Fourier recursion completes the proposed discrepancy test and shows why ordinary equidistribution points in the wrong direction: it predicts exponentially many noncrossing starts below $(3/2)^r$, whereas the proof would need exceptional one-sided avoidance of that entire initial interval. The finite spectrum exhibits short-horizon dispersion but supplies no global rigidity theorem. The adelic generating-function theorem excludes every permanently subcritical word with rational weighted generating function, including eventually periodic words, and the band-growth theorem excludes every

infinite word with $h_r = o(\log r)$. The d-kernel analysis shows why the remaining aperiodic corridor cannot simply be declared automatic: even the fixed positive orbit has infinite-dimensional kernels, and locally admissible aperiodic words occur in both regular and nonregular forms. Any regular counterexample model would be trapped in a logarithmic excess strip, but no theorem forces an ordinary orbit into that class. These are genuine aperiodic restrictions, not a closure of the corridor. The large-bit calculations make the pointwise obstruction more concrete. The geometric valuation law repels Haar-random starts from the noncrossing corridor at an exponential rate, while $2^B - 1$ and $2^B - 5$ show how positive truncations can shadow negative 2-adic periodic points for a number of steps proportional to their bit length. The exact equivalence between exclusion of an infinite positive corridor and L(r) tending to infinity separates the proved density decay from the missing least-representative growth theorem. The extended exact scan certifies L(r) through r = 344 and proves L(345) > 99,999,999,999. The origin-sensitive calculation shows why this finite growth is not a routine consequence of Haar decay: the anchored maximum can be held up by a small cluster while the average density falls. The mixed-residue enumeration likewise shows strong coupling between small binary and ternary coordinates even when the ternary marginal has low discrepancy. Exact power-gap computation provides a safe first-crossing window through 200,000 steps for starts above 984,572,779,225, but it does not force the first crossing. A proof that every orbit eventually falls below its starting value would resolve the Collatz conjecture; no such proof follows from the binary scaling or positive-density arguments presented here.